

Categorical and Geometrical Methods in Physics

Yuri Ximenes Martins

July 15, 2026

Contents

I	Overview	3
1	Intuition	4
1.1	Math and Physics	4
1.2	Unification	8
1.3	Insight	12
2	Classical	17
2.1	Geometry	17
2.2	Theories	22
2.3	Motion	32
3	Abstraction	33
3.1	Fields	33
3.2	Assertion	41
3.3	Action	46
3.4	Covariance	47
4	Quantum	48
4.1	Theories	48
4.2	Structure	52
4.3	Classification	54
5	Quantization	57
5.1	Definition	57
5.2	Path Integral	61
5.3	Pull-Push	66

Introduction

- the text was written in a
- For a physicist, maybe only the first part will be interesting.
- For a mathematician, the first part can be seen as an introduction/motivation for the concepts that will be formalized and the theorems that will be demonstrated in the second and in the third parts. Therefore, the mathematician reader which do not need a brunch of motivations to study and to do mathematics can omit the first part without loss of logical consistence.
- ainda que se tenha dito que a primeira parte foi totalmente devota a motivações, não se deve esperar que as partes dois e três tenham sido escritas de maneira seca e contraintuitiva. Em verdade, nos esforçamos enormemente para que o oposto viesse a tona. De fato, na parte dois, ao desenvolvermos o ferramental categórico e geométrico, todas as definições são apresentadas de modo que não pareçam arbitrárias, mas sim partes naturais de um contínuo raciocínio. Por sua vez, na parte três apresentamos os conceitos físicos sempre tendo em mente obter uma abstração daquilo que foi desenvolvido na primeira parte.
- we initiate the third chapter showing that each important example of classical theory can be considered as being a sigma-model theory if we use certain abstract language to describe them. In the sequence we show that such different types of sigma-models are simply parts of a same cohomology theory with different coefficients.
- The prerequisites relies in algebraic and differential topology: the first two parts of [my book] (which in turn are almost self-contained) have all that we will need. On the other hand, for completeness, was introduced two appendix containing most of the structural definitions and results (without proofs) which will be used along the text.

Part I

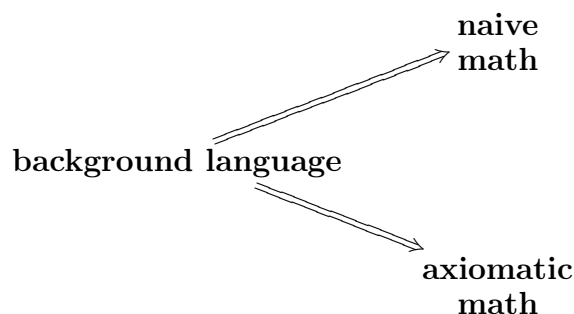
Overview

Chapter 1

Intuition

1.1 Math and Physics

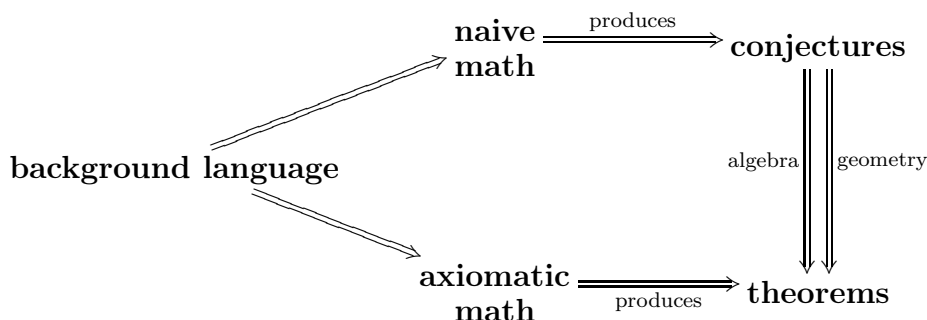
- There are two kinds of math: *naive* (or *intuitive*) *math* and *axiomatic* (or *rigorous*) *math*. In the naive math, the fundamental objects are assumed *primitives*. On the other hand, in the axiomatic math, these objects are defined by more elementary structures.
- **Example:** we have *naive set theory* and *axiomatic set theory*.
- independently of the case (naive or axiomatic) we need a *background language* (also called *logic*) to develop the theory.
- **Example:** in set theory, the language is *classical logic*.
- Thus, all that we have discussed can be summarized in the following diagram:



- Now we observe that when doing math, a mathematician does not make use of rigorous arguments. Indeed, is only after make intuitions and reflections about some result that he uses of the all rigorous necessary to valid it. So, we can say that naive math produces conjectures and rigorous math produces theorems. Furthermore, naive math comes before than axiomatic math. With effect, to pass from naive

math to rigorous math is necessary fix a model in which the conjecture is then proven.

- In general, we can fix *algebraic* models or *geometric models*, which reflect the existence of a duality between algebra and geometry. So, there is a question about uniqueness: *the choose of algebraic or geometric models produces equivalent results?* We will see examples of such duality in many moments of the text. In some of then, the previous question is actually an open question.



Categories

- There is a natural language when we need develop a theory that makes connections between among different disciplines of math: the *categorical language*.
- Giving a *category* is just the same that specifying an area of math. More precisely, a category is an entity composed of three kinds of information: *objects*, *mappings* (also called *morphisms*) between objects and an associative *composition* of mappings. We also require the existence of *identity morphisms*. So, a category theory is just an entity in that we can talk about *commutative diagrams*.
- **Example:** there is a category **Set**, whose objects are sets, whose morphisms are exactly functions between sets and composition is the usual. The are of math specified by **Set** is just *set theory*.
- **Example:** the most of the categories that we will study are *concretes*. This means that objects are sets with additional structures, morphisms are functions between the underlying sets preserving additional structures and composition is usual.
- **Example:** we have *algebraic categories* (generically denoted by **Alg**). Objects are sets with additional operations and morphisms are functions that preserve this operations (is this case, they are called *homomorphisms*). So, we have the category **Grp** of groups, the category **Rng** of rings, the category **Mod_R** of *R*-modules, the category **Vec_ℝ** of vectors spaces, etc. The correspondents areas of math are group theory, ring theory, module theory and linear algebra.

- **Example:** we also have a concrete category **Top** that corresponds to the area of *topology*. Indeed, your objects are topological spaces and your maps are continuous functions. To see that this category is concrete, remember that a topological space is just a set with an additional structure: a qualitative notion of *proximity of points*. Furthermore, a function between topological spaces are continuous exactly when preserve the correspondent notions of qualitative proximity.
- **Example:** **Diff** also specify an area of math: *analysis*.
- **Example:** there are categories which are not concrete. An important example is **nCob**. Their objects are manifolds without boundary of dimension n . A morphism $\Sigma : M \rightarrow M'$ is a *cobordism* between M and M' . That is, a $(n + 1)$ -dimensional manifold Σ such that $\partial\Sigma = M \sqcup M'$. Given

$$\Sigma : M \rightarrow M' \quad \text{and} \quad \Sigma' : M' \rightarrow M''$$

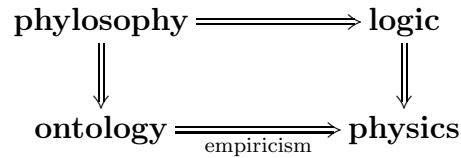
your composition must be a morphism $\Sigma' \circ \Sigma : M \rightarrow M''$. The idea is put $\Sigma' \circ \Sigma$ being the manifold obtained gluing Σ and Σ' along the boundary component M' . To be a category, we need identity morphisms $id_M : M \rightarrow M$ and the associativity of the composition. For the existence of identities, enough put $id_M : M \rightarrow M$ as being the *trivial cobordism* $id_M = M \times I$.

- *****IMAGE GLUING MANIFOLDS*****
- Now, to see that the categorical language is really natural to talk about connections between areas of math, observe that the action “to specify an area of math” is itself an area of math!! This means that we can talk about mappings between different categories (and therefore, between different areas of math). This mappings are called *functors*. On another words, there is a category **Cat**, whose objects are categories and whose morphisms are functors. The area of math delimited by **Cat** is just *category theory*.
- But, what is a functor? To define it, remember that in general a mapping is a kind of rule that preserve all structures. A category is an entity with objects, morphisms, composition and identities. So, a functor between **C** and **D** just be a rule $F : \mathbf{C} \rightarrow \mathbf{D}$ that take objects and morphisms of **C** and return objects and morphisms of **D**. Furthermore, F must satisfy $F(g \circ f) = F(g) \circ F(f)$ and $F(id_X) = id_{F(X)}$.

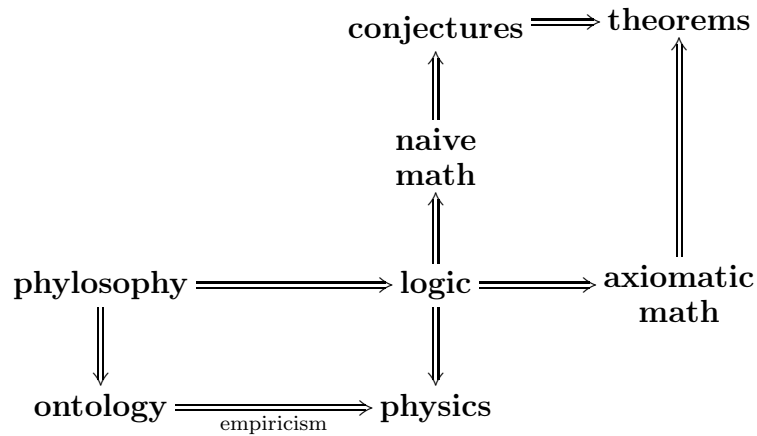
Physics

- In physics, the situation is totally different. Indeed, in physics we have a restriction about the *existence*. More precisely, we have a connection with *ontology*. The connection is given by the *empiricism*. So, more precisely, *although math is a strictly logical discipline, physics is logical and ontological*.

- This restriction produces many difficulties. For example, to consider a sentence as true, the logical consistence is not sufficient: we also need ontological consistence. So, *even tough a sentence be logical consistent, to be considered true it must be consistent with all possible experiment!* This also can be expressed in terms of a commutative diagram:



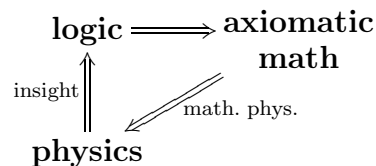
- But, remember that logic determines the diagram of math. So, gluing the diagram of physics with the diagram of math, we have a new commutative diagram:



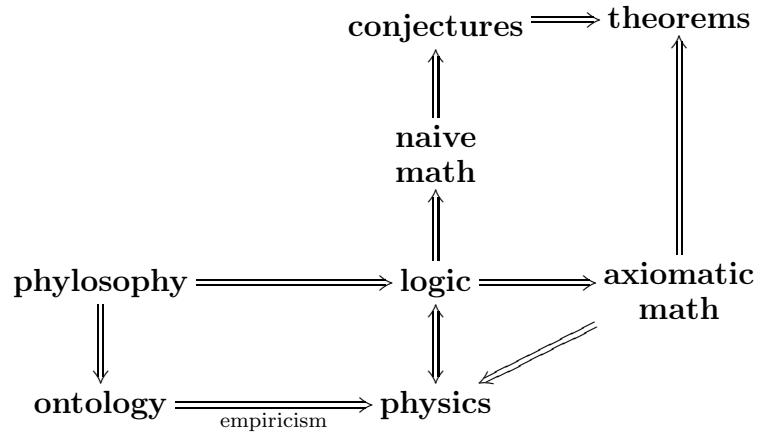
- the most important fact that concerns the relation between physics and mathematics is the following: in the previous diagram, the arrow

$$\text{logic} \Longrightarrow \text{physics} \quad \text{has an inverse} \quad \text{physics} \Longrightarrow \text{logic}$$

It is called *physical insight*. So, the composition of *physical insight* with axiomatic math produces a new arrow called *mathematical physics*.



- The full relation between physics and mathematics is thus given by



- **Conclusion:** *we can use physical insight to do naive math and, therefore, to produce conjectures. These conjectures can eventually proven, getting theorems!!! Furthermore, with the gotten theorems we can create mathematical models to physics (mathematical physics).*
- Now, we can say in simple terms what is the major objective of this text: *study some examples of the following sequence:*

$$\text{physics} \implies \text{logic} \implies \begin{matrix} \text{naive} \\ \text{math} \end{matrix} \implies \text{conjectures} \implies \text{theorems}$$

1.2 Unification

- Let pay special attention in the relation between logic and physics. That is, in the arrow

$$\text{logic} \implies \text{physics}$$

- consider now, the following objectives:

- unify math
- unify physics.

- how math is determined only by logic, *to unify math we **need** a very general background language.* On the other hand, physics is partially determined by logic, of mode that *more abstract and general background language **help us** to unify physics.*
- **Question:** how construct more general logic? First we observe that the abstraction of *set theory* to *category theory* characterize the abstraction of *classical logic* to *categorical logic*. So, if in some sense we formalize the passage of set theory to

category theory, we can iterate the process to get more and more general languages, just like the below diagram

$$\begin{array}{ccccc} \text{set theory} & \Longrightarrow & \text{category theory} & \Longrightarrow & \text{more abstract theory} \Longrightarrow \dots \\ \text{(classical logic)} & & \text{(categorical logic)} & & \text{(more abstract logic)} \end{array}$$

- A set is composed of *only one kind* of information: their elements. On the other hand, a category is composed by *three kinds* of information: objects, morphisms between objects and composition of morphisms. Thus, the passage of set theory to category theory is characterized by “*addition degrees of categorical information*”. So we expect that the passage of category theory to another more general theory also be characterized by addition of degrees of categorical information.
- In other words, we expect that this more abstract theory have not only objects, morphisms and one composition, but also *morphisms between morphisms that can be composed of two different ways*. We call this entities of *2-categories*. Adding another degree categorical degree we get 3-categories, and so on. In particular, we call a set a *0-category*. So, the previous diagram can be translated as

$$\begin{array}{ccccc} \text{set theory} & \Longrightarrow & \text{category theory} & \Longrightarrow & \text{2-category theory} \Longrightarrow \dots \\ \text{(classical logic)} & & \text{(categorical logic)} & & \text{(2-categorical logic)} \end{array}$$

- A way to build “more categorial concepts” is by the *categorification process*. Before present it, let see some examples of 2-categories, etc.
- **Example:** taking the category of all 0-categories we get **Set**, that is a 1-category. Analogously, taking the category of all 1-categories we get **Cat** that we affirm be a 2-category. To see this, we need construct a notion of “mapping between functors” that can be composed of two different ways. Indeed, this notion corresponds to the concept of *natural transformations*, whose we can compose in the *vertical* and *horizontal* way. So, more generally, taking the category off all $(n - 1)$ -categories we get a n -category which will be denoted by **Cat_n**.
- **Example:** the category **Top** is, in fact, a ∞ -category. Indeed, a 2-morphism $H : f \Rightarrow g$ between maps $f, g : X \rightarrow Y$ is just a homotopy between them. So, **Top** is a 2-category. But remember that a homotopy is also a continuous map. So, we can talk about homotopies between homotopies!! This corresponds to the 3-morphisms. But we can also talk about homotopies between 3-homotopies, and so on.
- **Example:** for every topological space X we can associate a ∞ -category $\Pi(X)$, defined in the following way: your objects are elements of X . A morphism $f : x \rightarrow y$ is a path connecting x to y . A 2-morphism between paths connecting x and y is a homotopy between them. Higher morphisms are homotopies between homotopies.

- **Example:** the categories $\mathbf{Cob}_n$ is, in truth, ∞ -categories.
- **Example:** for a given n , we can also build the n -category $\mathbf{Cob}(n)$. It is essentially the *complementary part of* $\mathbf{Cob}_n$. This means that, if $\mathbf{Cob}_n$ starts with n -manifolds, on the other hand $\mathbf{Cob}(n)$ ends with them. More precisely, their objects are 0-manifolds, their morphisms are cobordisms between 1-manifolds and so on, ending with n -morphisms, which are cobordisms between $(n - 1)$ -manifolds.

Categorification

categorification. Example: categorification. various models to monoidal category. There are no reason for the equivalence between this models.

- $\mathbf{Cob}_n$ is a monoidal category with disjoint union.
- $\mathbf{Cob}(n + 1)$ can be view as the categorification (by enrichment) of $\mathbf{Cob}(n)$.

Weakening

- weak commutative diagrams. Coherence conditions, etc.

Presentations

- HOMOTOPY THEORY as formalization (or presentation) of HIGHER CATEGORY THEORY
- homotopy hypothesis

Categorification in Physics

- here, we finally explain how categorification can be important in the unification of physics. To do this, first we observe that in the modern abstract viewpoint there are two kind of physical theories: *classical theories* and *quantum theories*.
- both are based on the notion of *abstract motion* of a certain p -dimensional object, generically called a *p-brane*, along the time. Such motion is given by a manifold of dimension $p + 1$. When $p = 0$ we also say that we have a *particle* realizing *worldlines*. For $p = 1$, we say that they are *strings* describing *worldsheets*.
- In the most general aspects, a *classical theory* deals with possible ways to realize the abstract motion Σ in some ambient M . More precisely, they select some smooth maps $\varphi : \Sigma \rightarrow M$, called *configurations*. In general, it is done giving a functional $S : \text{Map}(\Sigma; M) \rightarrow \mathbb{R}$ and looking for the configurations that minimizes S . On the other hand, *quantum theories* are ways to represent the abstract motions in terms of vector spaces and linear transformations.

- We will discuss classical and quantum theories in more detail later. At this moment, we focus on the abstract motions.
- **Example:** in the $n = 0$ case, Σ must be a discrete set and, therefore, a collection of points. They correspond, exactly, to the abstract motion of 0-dimensional objects (like particles) which not evolve in the time.
- **Example:** in the $n = 1$ case, the worldline Σ may have or not boundary. In both case, every connect component of Σ (say Σ_i) is determined by the classification theorem of one dimensional manifolds. Indeed, if such component is compact, then it is $\mathbb{S}^1$ or $I = [0, 1]$. On the other hand, if Σ is non compact, then Σ_i must be $\mathbb{R}$ or some interval like $(0, 1]$ or $[0, 1)$. In any situation, each component describes the motion of a particles. Indeed,
 - if $\Sigma_i \simeq \mathbb{S}^1$, then we have a generic *periodic motion*. Configurations $\varphi : \Sigma_i \rightarrow M$ are just *loops* on M ;
 - If $\Sigma_i \simeq \mathbb{R}$, we have an abstract *eternal motion*. In the other cases, we have particles moving along a certain interval of time. In both situations, configurations are *paths* on M .
- For $n > 1$, we don't have a complete classification of n -manifolds. Nevertheless, *there is a natural class of abstract motions Σ for each n* . Indeed, let look for the previous example. For the case of generic 0-objects moving along the interval $[0, 1]$, the manifold Σ is just a *cobordism* between the “initial” and “final” positions of the particles. We observe that this also happens with the other cases. Indeed, an interval like $(0, 1]$ or $[0, 1)$ is just a cobordism between a point (0 or 1) and the empty manifold $\emptyset$. Furthermore, $\mathbb{R}$ and $\mathbb{S}^1$ can be view as cobordisms between $\emptyset$ and itself. So, *every worldline of a particle is a cobordism*.
- For a given n , we can look analogously for the class of abstract motions Σ given by cobordisms. Indeed, if $\partial\Sigma = \Sigma_0 \sqcup \Sigma_1$, then Σ describes the worldvolume of a generic $(n - 1)$ -object with starts in Σ_0 and ends in Σ_1 . But certain morphisms of $\mathbf{Cob}_n$ can be view as n -morphisms of $\mathbf{Cob}(n)$.
- Now we can see how categorification is useful in the unification problem of physics. With effect, remembering that $\mathbf{Cob}(n + 1)$ is just the categorification of $\mathbf{Cob}(n)$, we conclude that *categorifying a theory of p -branes we get a theory of $(p + 1)$ -branes*.
- **Example:** as we have seen, $\mathbf{Cob}(1)$ describes *all possible* abstract motion of a particle moving along the time. Categorifying, we get a 2-category $\mathbf{Cob}(2)$, which describes not only particles, but *certain* abstract motions of strings. These motions are just ones which are cobordisms between 1-manifolds. As we will discuss later, this means that the string theories obtained by categorification is strictly *pertubative*.

1.3 Insight

- In the two previous subsections, we given special attention to the consequences produced in physics by changes in logic. Now, we will pay attention on the reciprocal: we will look to the arrow

$$\text{physics} \Longrightarrow \text{logic} .$$

- More precisely, we will look to situations in which physical insight can be effectively used to do naive math and, therefore, to produce conjectures. So, we will look to the following sequence

$$\text{physics} \Longrightarrow \text{logic} \Longrightarrow \begin{matrix} \text{naive} \\ \text{math} \end{matrix} \Longrightarrow \text{conjectures} .$$

- We will give three examples. They produces conjectures about the existence of higher nontrivial isomorphisms between different areas of math. The first example is the *mirror symmetry*. The second is the *holographic principle* and the third is about the existence of a *M-theory*.

Mirror Symmetry

- Remember that the categorification process take a concept and return a more abstract version of then. So, categorification produces new concepts, which can be realized by many examples. This realizations are, in principle, *totally independents*.
- For example, categoryfing the classical concept of *monoid* we get the abstract notion of *monoidal category*. Entities realizing this notion are $\mathbf{Mod}_R$ (with tensor product) and $\mathbf{Cob}_n$ (with disjoint union), which are not equivalents.
- Categorifying a theory of p -branes, we get a theory of $(p+1)$ -branes. In particular, categorifying theory of particles we obtain theory of strings. So, similarly, we expect that there are many ways to realize the concept of string theory, each of them living in a different area of math. In principle, these realizations need not be equivalents. But, by the physical point of view, they will describe the same kind of motion. So, we expect that they will produce the same physical predictions.
- In other words, the physical insight about the equivalence of physical predictions allow us to enunciate the following conjecture:
- **conjecture:** *the different realizations of the abstract concept of string theory (and, therefore, the different areas of math which describes them) are related in a nontrivial way.*

- We will call these nontrivial relations generically by *mirror symmetry*. So, we are looking for the following sequence:

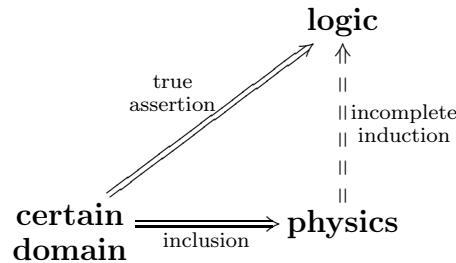
$$\begin{array}{ccccccc} \text{physics} & & & & & & \\ \text{uniqueness} & \implies & \text{logic} & \implies & \text{naive} & \implies & \text{conjectures} \\ \text{predictions} & & & & \text{math} & & \text{mirror symmetry} \end{array} .$$

- As will be discussed along the text (specially in the final chapter) there are to many five kinds of string theories, which are really connected by certain kind of nontrivial isomorphisms called *T-duality* and *S-duality*. Such theories are called Type I, Type II A, Type II B, $SO(32)$ Heterotic and, finally, $E_8 \times E_8$ Heterotic. The relations between them are represented in the below diagram.

- *****DIAGRAM*****

Holographic Principle

- In physics, *every inductive argument is falsifiable*. So, we cannot use the validity of some particular cases to conclude the validity of the general case. On the other hand, if a certain kind of affirmation agree with many reliable experiments, we can say that it is *partially true* (or that is *true in a certain domain*). So, the expectancy about the extrapolation of the validity affirmation for every experiment creates conjectures.



- In more precise terms, how in physics we cannot get theorems by the induction method, the usage of (incomplete) induction will produce conjectures. That is, incomplete induction provides the extension represented in the above diagram, allowing us to examples of the sequence:

$$\begin{array}{ccccccc} \text{physics} & & & & & & \\ \text{incomplete} & \implies & \text{logic} & \implies & \text{naive} & \implies & \text{conjectures} \\ \text{induction} & & & & \text{math} & & \end{array} .$$

- **Example:** There are some examples of physical systems which are described by *fields* defined on a $(n + 1)$ -manifold Σ , but with behavior on the boundary $\partial\Sigma$ totally determined by certain *source*. Indeed,

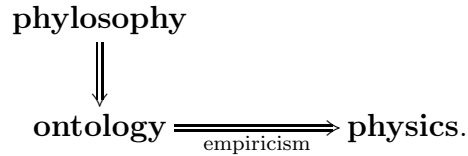
- the Gauss law of electrostatic say that the flow of the electric field along the boundary of a closed surface is determined by the distribution of electric charges inside the surface;
 - the Ampère’s law of magnetism say that the magnetic field on the circuit delimited by the boundary of an open surface is determined by the distribution of current which cross the surface;
 - the entropy of a black hole is determined by the area of the boundary of their horizon of events.
- So, in certain domain, the affirmation that any physical theory on Σ is determined on the boundary $\partial\Sigma$ by another theory is true. We say that the theory on Σ is the *bulk theory* and that the theory in $\partial\Sigma$ is the *source theory*. We also say that they are *holographic duals*. Therefore, applying incomplete induction, we get the following conjecture, called *holographic principle*.
 - **conjecture:** *any physical theory of p -branes (and, therefore, any area of math describing it) has some holographic dual theory of $(p - 1)$ -branes.*
 - In particular, observe that this conjecture give a correspondence between theories of particles and theories of strings. As we will discuss later, the classical situation is given by the Ads-CFT duality, in which certain gravitational theories (described as string theories) can be described as (gauge conformal) theories of particles.
 - POISSON-SIGMA model (classical mechanics view as string theory). By holographic principle, corresponds a quantum field theory, which is expect to be exactly quantum mechanics.

Perturbative and Full Theories

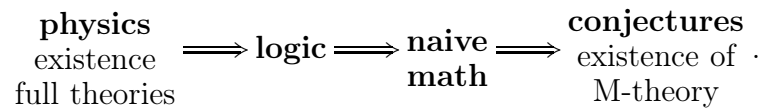
- By the physical viewpoint, there are two kinds of interactions: *external* and *internal*.
- **Example:** we can consider the earth subjected by an external potential like $-\frac{GM}{r}$ or we can look for the system (earth+sun) interacting by the Newton’s universal law of gravity. The first is an example of *external* interaction, as well the second is an *internal* interaction.
- We observe that, *in the given definition of abstract motion of particles, only external interactions was considered*. Indeed, typical examples of internal interactions are *scatterings*, which can be described, for example, by *collisions*. In such type of interaction, we have a *vertex*, like the present below. So, differently that we have supposed, abstract motions describing scatterings generally *are not manifolds*, but a certain kind of *graph* (usually called *Feynman graph*).
- ***** FEYNMAN GRAPHS*****

- In the previous example, both descriptions are completely equivalents. More, precisely, in the previous example we pass from external to internal interaction simply including the source of the interaction on the system. On the other hand, we can pass from internal to external description simply isolating some of the bodies (earth or sun).
- So, we expect that the same must be valid for an arbitrary particle system. That is, we expect that every physical theory (being classical or quantum) can be described in an equivalent way using external or internal interactions. In other words, considering the configurations as embedding manifolds on the target or as graphs having certain vertex.
- The external perspective (which was used until the present moment) is called *full* or (*non perturbative*). The internal, on the other hand, is called *perturbative perspective*. So, analogously to the non perturbative case, in the perturbative case, classical theories of particles are mechanisms which select the possible graphs, as well quantum theories of particles are mechanisms which assign certain probability to every graph.
- We are looking for a generalization of the perturbative perspective for theories of p -branes, with $p > 1$. Observe that, for $p > 1$, we have no vertex of interaction, of mode that the “higher dimensional graphs” also are manifolds. This means that our definition of abstract motion of p -brane, with $p > 1$, must include the perturbative perspective which we are searching.
- Now, remember that abstract motions of p -branes given by cobordisms (that is, by the morphisms of $\mathbf{Cob}(p)$) are naturally distinguished. This motivates to consider
 - *perturbative theory* of p -branes as being ones which takes into account only the worldvolumes given by cobordisms. Furthermore, when $p = 1$, they describes all possible abstract motions;
 - *full theory* of p -branes as being ones which takes into account all worldvolumes.
- The above definitions clarify two things:
 - in principle, *non perturbative theories can describe full information* about the motion of p -branes, as well *perturbative theories can describes only partial information*.
 - every full theory induces, by restriction, a perturbative theory.
- In the particular case when $p = 1$, every motion is a cobordism and then every information can be reconstructed by a perturbative theory, reflecting the equivalence between the perturbative and non perturbative prescriptions. So, *the previous definitions really generalizes the given for particles*.

- Now, remember the connection between physics and ontology. That is, remember the existence of the arrows



- So, a certain kind of information exist iff can be accessed using some experiment. Consequently, *we expect that every perturbative theory is the restriction of some full theory*. Otherwise, there should be a perturbative theory describing full information, which is a contradiction.
- **Example:** categorifying a full theory of p -branes we get a perturbative theory of $(p + 1)$ -branes. This theory must be the restriction of some full theory. So, in particular, must be exists a full theory of strings, which is usually called *M-theory*.



- EFFECTIVE THEORIES.
- IN GENERAL, GRAVITY IS NON-RENORMALIZABLE. WE EXPECTED THAT IT IS THE EFFECTIVE THEORY OF SOME OTHER THEORY.
- SIMILARLY, THE STANDARD MODEL WORKS ONLY IF NEUTRINOS ARE MASSLESS. EXPERIMENTS SHOW THE OPPOSITE, SO STANDARD MODEL IS EFFECTIVE THEORY OF SOMETHING.

Chapter 2

Classical

- We have seen that physical insights produce conjectures. Now, remember that to turn conjectures in theorems, we need fix some model. In general, these models are given by certain kind of *geometric framework*. In this section, we will give some examples of these realizations.

$$\text{physics} \implies \text{logic} \implies \begin{matrix} \text{naive} \\ \text{math} \end{matrix} \implies \text{conjectures} \xRightarrow{\text{geometry}} \text{theorems}$$

- As we will see, they appear tom many in four kinds of situations:
 - in the description of *external interactions*;
 - in the description of the *force* produced by some interactions;
 - in the description of entities with *additional degrees of freedom*;
 - in the *quantization process*.
- Now, we will discuss the three first situations. The geometric quantization process will be described in the next chapter.

2.1 Geometry

- Remember that there are two kinds of physical theories: *classical* and *quantum*. In a classical physical theory, we work with embeds $\varphi : \Sigma \rightarrow M$ of abstract motions Σ on a target space M . That is, we work with movements of entities in a certain ambient. This is the notion of *configuration*.
- Now, we can ask: *what produces these movements Σ* ? The answer is: *the external interactions with act over the entity while it moves in M* . Now, observe that such interaction (being external) must be described by properties of the ambient M and not of the worldvolume Σ or of the configuration φ . So, generally we can say that *external interactions are induced by additional structures over M* .

- In terms of diagrams, let represent certain additional structure on M as a mapping $\pi : P \rightarrow M$, which will be called *bundle* over M . We call $P_p = \pi^{-1}(p)$ the *structure* (or *fiber*) on p . So, the configurations of a physical theory over M is given by diagrams like this:

$$\begin{array}{ccc} & P & \\ & \downarrow \pi & \\ \Sigma & \xrightarrow{\varphi} & M \end{array}$$

- Now, observe that the interactions must act on a certain region U of the space and must be dependent of the structure P . So, they are given by maps $s : U \rightarrow P$, such that $s(p) \in P_p$. That is, such that $s \circ \pi = id_U$. They are called *local sections* for the bundle π and are the *background fields of the interaction*.
- The idea is the following: the bundle defines the *nature* of the interaction, as well the sections are the entities with *mediate* the interaction. Furthermore, if $\varphi : \Sigma \rightarrow M$ is a configuration, then $s \circ \varphi$ describes the resultant configuration subjected to the interaction mediated by s .
- **Example:** tangent bundle and vector fields.
- In many cases, the bundle $\pi : P \rightarrow M$ is endowed with an (free and transitive action) action of a Lie group G on P . In these situations, we say that we have a *G -principal bundle* over M . About this kind of bundle, let do some observations:
 - the most simple G -principal bundle, called *trivial*, is ones in which $P = M \times G$ and G act by left translation;
 - we can show that a principal bundle has a global section iff is equivalent to the trivial bundle. So, *the interactions described by a nontrivial principal bundle is always local*.
- A natural question is on the importance of G in the structure of the principal bundle: *is always possible change G by another Lie group H* ? About this, observe that every $H \rightarrow G$ induces an action over P . In general, this new mapping is *not* a principal bundle over M . Indeed, there are certain *topological obstructions*, usually given by *characteristic classes*, for that it be.
- A particular situation happens when $H \rightarrow G$ is the inclusion. In this case, the H -principal bundles over M are in bijection with the sections of $G/H \rightarrow M$. But, the quotient G/H is essentially *geometry in the context of Klein*.
- Therefore, we can conclude that: *nothing all interactions can be described in a fixed ambient and the problem about the possibility of describe some of them is a geometric problem with pass from topological obstructions*.

- **Example:** *G-structures*. Example: orientation. The obstruction is the so called *first Stiefel-Whitney* class $w_1(M)$. $SO(1, n - 1)$ entities moving in a gravitational ambient. For $p = 0$ and $n = 4$, as we will see, this can be used to describe the *gravitational interaction*.
- EXAMPLES OF GEOMETRIES, THEIR TOPOLOGICAL OBSTRUCTION AND THEIR INTEGRABILITY CONDITIONS.

Force

- In the last subsection, we have seen that principal bundles are the objects which generally describe external interactions. Indeed, the interactions are mediated by the sections $s : P \rightarrow M$.
- Remember, now, that physics is an empirical discipline. So, we need detect experimentally the interaction which the entities are subjected. But, observe that *we detect only the local behavior of the interaction*, what corresponds to the *forces*. With this in mint, in this subsection we will see that the principal bundles also are the natural ambient in which we can talk about *forces acting on particles*.
- The idea is the following: let consider a theory which describes a **particle** moving an ambient subjected to certain interaction. This interaction (which is given by a certain section) act on every point (and, therefore, over every configuration $\varphi : \Sigma \rightarrow M$ of the particle) in the target space. Suppose that we have a way to *transport information along paths*. So, along the transportation, the interaction act on the configurations $\varphi : \Sigma \rightarrow M$. Therefore, looking for the difference between the information at initial instant and at the final instant, we can have a local idea of the behavior of the interaction, which will be the required notion of force.
- Let see that this notion of transportation along paths (and, therefore, the notion of force) really can be introduced on every (not necessary principal) bundle.
- If $f : P \rightarrow M$ is any map, then the kernel of the derivative Df_a determines, in every point $a \in P$, a distinguished part of TP_a and, therefore, defines the *subbundle* $V \subset TP$ of *vertical vectors*. Now, suppose that we have a complementary subbundle of *horizontal vectors* H such that $V \oplus H = TP$.
- Then we can talk about path that realizes *horizontal transportation*. Indeed, this will be exactly the configurations for which every $\varphi'(t)$ is a horizontal vector. So, to measure locally certain interactions, $\varphi : \Sigma \rightarrow M$ and analyses if the horizontal vectors $\varphi'(0)$ and $\varphi'(1)$ coincides.
- In the case of G -principal bundles, the situation is more concrete:

- *we can identify the notions of horizontal transportation.* Indeed, they are in bijection to certain 1-forms with values in the Lie algebra $\mathfrak{g}$, called *connections* on P (equivalently, such forms are given 1-forms $A_i : U_i \rightarrow \mathfrak{g}$ over a cover of M , which are compatible in the intersections $U_i \cap U_j$);
- *for every $a \in P$ and for a fixed connection A , we have a way to quantify the failure of horizontal transportation along loops with start and end in a .* Indeed, it is given by the subgroup $\text{Hol}(A; a) \subset G$ of every g for which there is a horizontal path $\varphi : a \rightarrow g * a$.
- Furthermore, each connection A determines a modification on the exterior derivative d turning it *covariant*. In this case, it is denoted by d_A . Indeed, $d_A \omega = \text{hor}(d\omega)$. The covariant 2-form $\Omega = d_A A$ is called *curvature* of the connection A . Differently what happens with d , we don't have $d_A^2 = 0$. Indeed, we can compute $d_A^2 \omega = \Omega \wedge \omega$, of mode that the *curvature is exactly the obstruction to $d_A^2 = 0$* .
- We also have a result, called *Ambrose-Singer theorem* which states that the lie algebra $\mathfrak{hol}(A, a)$ of the holonomy group $\text{Hol}(A; a)$ is generated by the image of the correspondent curvature Ω of A .
- **Conclusion:** *on principal bundles, holonomy describe locally the interactions over particles and curvature describes locally the holonomy. But curvature is the derivative of the connection. Therefore, force is essentially curvature and the connection is a certain kind of potential.*

$$\begin{array}{ccccccc} \text{connection} & \Longrightarrow & \text{curvature} & \Longrightarrow & \text{holonomy} & \Longrightarrow & \text{measurement} \\ \text{(potential)} & & \text{(force)} & & & & \end{array}$$

- How forces describes the local behavior of the interaction and how, for the case of particles, forces and described by connections, we expect that there are bundles whose sections are precisely connections. Indeed, this is the case: for every Lie group G and for every smooth manifold M , there is a bundle $\pi : G_{\text{con}} \rightarrow M$ whose (locally compatible) sections are just connections of G -principal bundles over M . The interactions described in this way are called *gauge interactions* and the group G is the *gauge group*.
- We finalize this subsection listing some problems on the gauge interaction discussed here:
 - first of all, the concept of holonomy (and, therefore, the notions of connection, curvature and force) are defined exclusively for theories of **particles**. We don't have a direct analogous for strings or more abstract objects;
 - in general, the bundle of connections of a given G -principal bundle over the target space M is not *principal*.
- The two problem can be solved passing to *higher geometry*, as will be discussed later.

Gravity

- Now, remember that physics is an empiricist discipline. So, how the interactions given by forces are exactly which can be measured, we expect that *any fundamental interaction of the nature which act on **particles** be a gauge interaction.*
- Actually, there are four fundamental interactions over particles: electromagnetic, weak, strong and gravitational. The first three are actually described as gauge interactions with gauge groups $U(1)$, $SU(2)$ and $SU(3)$, respectively. Together, they constitute the so called *Standard Model* for particles.
- On the other hand, gravitational interaction is generally described by certain G -structures which, in principle, *not define gauge interactions*. More precisely, remember that gauge interactions consider all connections taking values into a lie algebra $\mathfrak{g}$, while gravity work with all metrics. Happens that any metric induce many connections: such ones that are *compatible* with the given metric. So, there is a map like the following:
- *****
- So, the fact that classical gravity is not a gauge theory provides from the nonexistence of an inverse for the previous map. After all, there are connections which are not compatible with any metric.
- But observe that, although gravity is not a gauge interaction, it can be seen as a *partially gauge interaction* in the sense that rather consider all connections, we look only for which are compatible with some metric.
- More generally, we can consider an interaction described by any $H \hookrightarrow G$ reduction and look it as a partially gauge interaction for the connections $A \in \Omega^1(P; \mathfrak{g})$ which, in some sense, are *compatible* with the reduction. A way to describe this compatibility is obtained observing that the reduction $H \hookrightarrow G$ induces a look connections are called *Cartan connections* for $H \hookrightarrow G$.
- ***** ARRUMAR*****
- More precisely, fixed a connection A on a G -principal bundle $\pi : P \rightarrow M$, we say that a reduction to $H \subset G$ is a *Cartan reduction* (and, in this case, A is called a *cartan Connection*) when $A : P \rightarrow \mathfrak{g}$ pass isomorphically to the quotient $\mathfrak{g}/\mathfrak{h}$.
- **Conclusion:** *gravitational interaction over particles is naturally described by Cartan geometry on the reduction $O(n, 1) \subset \text{Iso}(n, 1)$.*

gravity for $\longleftrightarrow$ Cartan
particles geometry

- OTHER APPROACH TO GRAVITY: GRAVITATIONAL EFFECT CAN BE MEASURED BY THE NONINTEGRABILITY OF CERTAIN GEOMETRIC STRUCTURE.

Internal Degrees of Freedom

- We can also think about physical theories in which the entities whose movement we want describe have certain *additional internal degrees of freedom*. This is essentially the dual situation with respect to the previous. Indeed, here we are dealing with conditions *over the entity* and not over the ambient in which it will be moved.
- So, is natural consider bundles over a fixed abstract motion Σ , like this:

$$\begin{array}{ccc} P & & \\ \pi \downarrow & & \\ \Sigma & \xrightarrow{\varphi} & M \end{array}$$

- The sections $\psi : \Sigma \rightarrow P$ of this bundle will describe the possible *internal configurations* of the entity. Like as the previous situation, in general P corresponds to a certain kind of *geometric framework subjected to topological obstructions*.
- **Example:** the principal example of internal degree of freedom is the *spin*. The bundles which describes these internal degrees are *spinor bundles*. They are build *pulling back* certain bundles over the target space. More precisely, let M be orientable. So, their frame bundle admits a reduction to special orthogonal group. Suppose that such reduction admits an extension to the universal cover of $SO(n)$, which will be denoted by $Spin(n)$. For example, $Spin(3) \simeq SU(2)$. Now, every linear representation ρ of the spin group on some linear space $\mathbb{S}$ defines a vector bundle over M , which will be denoted by $\mathbb{S}M$. So, for a fixed configuration $\varphi : \Sigma \rightarrow M$, we can complete the square in a universal way to get a bundle over Σ , like below. Their sections $\psi : \Sigma \rightarrow P$ are called *ρ -spinors*.

$$\begin{array}{ccc} P & \dashrightarrow & \mathbb{S}M \\ \downarrow & & \downarrow \pi \\ \Sigma & \xrightarrow{\varphi} & M \end{array}$$

2.2 Theories

- Generally, a classical physical theory is determined by a function $S : X \rightarrow \mathbb{R}$, defined on a certain set X , called *phase space* (or *space of field configurations*) of the theory. We call S the *action functional* of the theory. To build an action functional describing

a theory, we first consider it defined on a natural domain. This means that, if we need a theory of a object moving along the abstract motion Σ on the ambient M and subjected to certain interaction $P \rightarrow M$, we look for actions defined on the all configurations $\varphi : \Sigma \rightarrow M$ and on the all sections $s : M \rightarrow P$. That is, we look for maps

$$S : \text{Map}(\Sigma; M) \times \Gamma(P) \rightarrow \mathbb{R}.$$

- For a fixed action functional, we search for certain *symmetries*. That is, mappings which stay the action invariant. Suppose that we having identified a set Sym of symmetries. Then the action pass to the quotient:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{S} & \mathbb{R} \\ \pi \downarrow & \nearrow S' & \\ X/\text{Sym} & & \end{array}$$

- GAUGE FIXING PROCEDURE.
- GROUPOIDS
- HIGHER SYMMETRIES.....
- **Example:** For a given Lie group G , let take an action of the form $S : G\text{Strc} \rightarrow \mathbb{R}$, whose space of configurations is the set of all G -structures over some manifold. Any of such functions pass to the quotient by the action of the automorphism group of such G -structures.
- **Example:** Extensions of a given symmetry. Supersymmetry as extension of the Poincaré group.
- Compactified theories.
- **Example:** *Calabi-Yau threefolds as the “compactified” part of string theory.* Our mensurable universe is $n = 4$. So, we need a split $M \simeq M^4 \times C^6$, where C^6 is a “highly compactified” semi-riemannian 6-manifold. There is a theorem (Berger’s classification theorem) that classify all possible geometries compatible with a given semi-riemannian geometry. This classification in given in terms of the *holonomy* and *dimension*. In dimension six, the holonomy is $U(3)$ or $SU(3)$, which corresponds to Kähler and Calabi-Yau geometries. But if a manifold have conserved spinors, so it is Ricci-flat. Thus, C^6 have exactly the structure of Calabi-Yau threefold.
- Physics is empiric. LHC does not observe global supersymmetry. So, the previous assumption is incorrect.

Conformal

There is a very special class of physical theories, called *conformal theories*. They are too special that we will look to them in an exclusive subsection. We start considering a theory of an entity of motion Σ moving into M . Now, let is suppose that Σ becomes endowed with metrics of certain assignature. So, an action functional for such movements is generally given by

$$S : \text{Map}(\Sigma; M) \times \text{Met}(\Sigma) \rightarrow \mathbb{R}.$$

We have two groups which act naturally on the set of metrics: $\text{Diff}(\Sigma)$ acting by pullback (as was discussed previously) and $C^\infty(\Sigma)$ acting by *rescalings*: $g \mapsto \Lambda \cdot g$, with $\Lambda = e^\lambda$. But each ones act on the another, producing a commutative diagram (we denote any of such final quotients by $\text{Conf}(\Sigma)$):

$$\begin{array}{ccc} C^\infty(\Sigma) \times \text{Met}(\Sigma)/\text{Diff}(\Sigma) & \longrightarrow & \text{Met}(\Sigma)/\text{Diff}(\Sigma)/C^\infty(\Sigma) \\ \uparrow & & \downarrow \simeq \\ C^\infty(\Sigma) \times \text{Diff}(\Sigma) \times \text{Met}(\Sigma) & & \\ \downarrow & & \\ \text{Diff}(\Sigma) \times \text{Met}(\Sigma)/C^\infty(\Sigma) & \longrightarrow & \text{Met}(\Sigma)/C^\infty(\Sigma)/\text{Diff}(\Sigma) \end{array}$$

A *conformal theory* is just one in which the action pass to the simultaneous quotients of both actions, being, therefore, defined on $\text{Conf}(\Sigma)$ instead of on $\text{Met}(\Sigma)$. In order to clarify what it means, recall that the elements of

$$\text{Met}(\Sigma)/\text{Diff}(\Sigma) \quad \text{and} \quad \text{Met}(\Sigma)/C^\infty(\Sigma)$$

are respectively classes of isometric metrics and *conformal structures*, i.e classes of metric metrics up to *rescalings factors*. Therefore, the elements of

$$\text{Met}(\Sigma)/\text{Diff}(\Sigma)/C^\infty(\Sigma) \simeq \text{Met}(\Sigma)/C^\infty(\Sigma)/\text{Diff}(\Sigma)$$

are classes of metrics which are isometric up to a rescaling or, equivalently, classes of conformal structures which are invariant by pullbacks. The diffeomorphisms of Σ which act on the metrics preserving the conformal structure are called *conformal transformations*. Together, they defines a subgroup of $\text{Diff}(\Sigma)$: the *conformal group*. So, we can say that *conformal theory* is simply one whose action functional is invariant by the action of the conformal group (i.e that is invariant by conformal transformations).

At this moment we can appreciate the (physical) importance of conformal theories: recall that one of the problems of physics is to build an universal theory which unity all the physical laws. Suppose that such theory exists. Different laws holds on different scales of energy, so that it is expected that this universal theory be invariant by rescalings, i.e it must be conformal.

Now, let us see that 2-dimensional conformal theories (which corresponds to conformal string theories) are further special. The reason, as will be explained below, is the following: the infinitesimal version of conformal transformations are the *conformal Killing vector fields*. When $\dim \Sigma > 2$ there are finitely many of such fields, but when $n = 2$ there are infinitely many of them.

The first step is to identify such conformal Killing fields, because integrating it we recover the conformal group. Recalling that a classical Killing field is one such that $\mathcal{L}_X g = 0$, we see that its conformal version are just those which satisfy $\mathcal{L}_X g = \Lambda \cdot g$ for some scale factor. Fixing a coordinate system, such equation writes

$$\nabla_{\partial_i} X_j + \nabla_{\partial_j} X_i = \Lambda(x_1, \dots, x_n) g_{ij}, \quad (2.1)$$

where ∇ is the Levi-Civita connection of g . Therefore, taking the trace with respect to g ,

$$2\operatorname{div}_g X = \nabla_{\partial_i} g^{ij} X_j + \nabla_{\partial_j} g^{ji} X_i = \Lambda g_{ij} g^{ij} = n\Lambda,$$

so that the unique possible conformal Killing fields for a given conformal factor e^λ are those that satisfy

$$\operatorname{div}_g X = \frac{n}{2} \Lambda, \quad (2.2)$$

clarifying that the number $n = 2$ is really special. Indeed, in such dimension the conformal Killing equation reduces to

$$\mathcal{L}_X g = \operatorname{div}_g X \cdot g.$$

We assert that there are infinitely many solutions of the above equation, so that in dimension equal to two the collection of conformal maps is really big. There are two signatures to analyze: Riemannian and Lorentzian. Independently of the case, the metric is always locally conformally flat see, for instance [??]). Indeed, we say that a semi-riemannian manifold is *conformally flat* when it is in the conformal class of a flat manifold. More generally, we say that it is *locally conformally flat* when it is conformally flat in the neighborhood of any point, i.e when for any $p \in M$ there is a chart $\varphi : U \rightarrow \mathbb{R}^n$, with $p \in U$, and a diffeo $f : U \rightarrow f(U)$ such that $\varphi^* g = \Lambda \cdot f^* h$ for some Λ , where h is a flat metric on $f(U)$. Therefore, in the Riemannian case the equation (2.1) is locally equivalent to

$$\partial_i X_j + \partial_j X_i = \Lambda(x_1, \dots, x_n) \delta_{ij}, \quad i, j = 1, 2,$$

which imply

$$\partial_1 X_1 = \frac{1}{2} \Lambda = \partial_2 X_2 \quad \text{and} \quad \partial_1 X_2 + \partial_2 X_1 = 0,$$

meaning that the functions X_i satisfy the Cauchy-Riemann and, in special, are part of a holomorphic map with nowhere vanishing derivative. Observe that, in such situation, by (2.2) we have

$$\partial_1 X_1$$

- similarly, every holomorphic X nowhere vanishing function defines an orientation preserving conformal Killing field with rescaling factor $\Lambda = |X'|^2$. Examples of those transformations are

$$X_n : \mathbb{C} - 0 \rightarrow \mathbb{C} - 0, \quad \text{given by} \quad X_n = -z^{n-1} \frac{\partial}{\partial z},$$

which generates an infinite dimensional Lie algebra.

- the Lorentzian signature admits a similar formulation which can be found in details in the (very nice) reference [??].

Now, recall that one of the main problems with a given classical theory is that its phase space generally do not has some finite dimensional structure. So, the presence of symmetries is very appreciated, because it allow us to cut down pieces of the phase space, turning it more well behaved. In the previous discussion we conclude that in dimension equal to two, conformal transformations are generated by infinitely many conformal Killing fields. Therefore, if a theory of strings admit conformal symmetry, then we can cut down a large part of the phase space, showing that such theories tends to be very well behaved. Indeed, when $n = 2$ the space $\text{Met}(\Sigma)/\text{Conf}(\Sigma)$ of (oriented) Riemannian structures module (orientation preserving) conformal maps is finite dimensional!

The idea is the following: the previous discussion suggest that in dimension equal to two there is a strict relation between Riemannian conformal structures and complex structures. With effect, as will be explained below, *oriented metrics up to (orientation preserving conformal maps) are just the same that complex structures up to biholomorphisms*. More precisely, we can prove that we have a 1-1 correspondence

$$\text{Met}(\Sigma)/\text{Conf}(\Sigma) \simeq \text{Complex}(\Sigma)/\text{Diff}(\Sigma). \quad (2.3)$$

But the right side is exactly the moduli space of complex structures, which is an orbifold (that is, an entity similar to a manifold, but instead of be modeled over $\mathbb{R}^n$, it is modeled over some quotient space $\mathbb{R}^n/\Gamma$ by the action of some discrete group) of dimension equal to $6g - 6 + 2k$, where g and k are the genus and the number of components at the boundary of Σ . For a proof of this fact, see e.g [??].

Let is sketch (2.3). We will proof that $\text{Met}(\Sigma) \simeq \text{Complex}(\Sigma)$. But the previous discussion suggest exactly that conformal maps are in 1-1 correspondence with holomorphic maps, giving the required result. We initiate observing $SO(2) \simeq U(1)$ so that to give an oriented structure is just the same that giving an almost hermitean structure (and, therefore, just the same that giving an almost complex structure, because $U(n) \subset GL(n; \mathbb{C})$ is the maximal compact subgroup). On the other hand, in dimension equal to two, the Nijenhuis tensor is equal to zero, so that oriented structures are equal to complex structures, ending the proof. Indeed, if $J : T\Sigma \rightarrow T\Sigma$ is the endomorphism generating the almost complex structure, then any vector field X is linearly independent to JX , such that for

any chart $x : U \rightarrow \mathbb{R}^2$, the pair $(\partial_1, J(\partial_1))$ defines a local basis. Therefore, being $J^2 = -1$

$$\begin{aligned} N_J(\partial_1, J\partial_1) &= [\partial_1, J\partial_1] + J[J\partial_1, J\partial_1] + J[\partial_1, J^2\partial_1] - [J\partial_1, J^2\partial_1] \\ &= [\partial_1, J\partial_1] + [J\partial_1, \partial_1] \\ &= 0, \end{aligned}$$

implying $N_J = 0$, because the Nijenhuis tensor is bilinear and alternated.

Concrete Examples

- Now, let give some specific examples of actions which actually describes interesting classical physical theories. All of them are *lagrangian theories* in the sense that the action $S : X \rightarrow \mathbb{R}$ Many of them are *local theories* in the sense that.....
- **Example:** *Bosonic Theories.*
- The most simple theories are ones in which the entity is to any interaction. Because of this, we say that it moves *freely* in the background ambient. For a theory of p -branes moving in a background ambient M (endowed with a fixed metric h), the action is given by the *Polyakov action*:

$$S_{Pol} : \text{Map}(\Sigma; M) \times \text{Met}(\Sigma) \rightarrow \mathbb{R}, \quad \text{with} \quad S[\varphi, g] = \int_{\Sigma} \|d\varphi\|^2 \cdot \omega_g,$$

where $d\varphi$ is the 1-form on $\Sigma \times M$ with values in φ^*TM and $\|\cdot\|$ is the norm induced on such objects by the metrics g on Σ and h on M . In local charts,

$$S_{Pol}[\varphi, g] = \int_M g_{ij} h_{kl} \partial_i \varphi^k \partial_j \varphi^l \sqrt{g} \cdot dx^1 \dots dx^n.$$

- Remembering that the choice of a metric g is equivalent to the choice of a $O(n)$ -structure, we see that S_{Pol} pass to the quotient $\text{Met}(\Sigma)/\text{Diff}(\Sigma)$. THIS THEORY IS CONFORMAL.
- bosonic theories with self-interaction.
- **Example:** *Fermionic theories.* We also have theories describing the free movement of p -branes which have internal degrees of freedom. The idea is the following: the fixed metric h on the ambient M is equivalent to Cartan geometry over $O(n)$ and then, therefore, induces a connection over TM . Pulling back such connection, we get a new connection on $\mathbb{S}M$, which pass (also pulling back) to P . The result is a $C^\infty(\Sigma)$ -linear derivation

$$D : \Gamma(P) \rightarrow \Gamma(P \otimes T\Sigma^*).$$

Now, suppose that the space $\mathbb{S}$ in which we are presenting $Spin(n)$ becomes endowed with a fixed linear map $\mathbb{S} \otimes (\mathbb{R}^n)^* \rightarrow \mathbb{S}$. It induces a morphism between the bundles $T\Sigma^* \otimes \mathbb{S}M \rightarrow \mathbb{S}M$. Pulling back by φ and applying Γ we get a morphism

$$\gamma : \Gamma(P \otimes T\Sigma^*) \rightarrow \Gamma(P).$$

The composition $\gamma \circ D$ is usually denoted by $\not{D}$. Now, to build the action (which is a real function), also suppose that $\mathbb{S}$ is endowed with a non-degenerated bilinear map represented by $(v, w) \mapsto \bar{v}w$. It induces a bilinear map on $\Gamma(P)$ with values in $C^\infty(\Sigma)$, and then we can put

$$S_{\not{D}} : \Gamma(P) \rightarrow \mathbb{R} \quad \text{as being} \quad S_{\not{D}}[\psi] = \int_{\Sigma} \bar{\psi}(\not{D}\psi) \cdot \omega_g, \quad \text{with} \quad g = \varphi^*h.$$

• **Example:** *Supersymmetric theories.* We

- the different types of string theories comment later are supersymmetric on the worldsheet.

• The previous theory is local supersymmetric on the worldvolume:

- fsdfsdfs

• **Example:** *RNS-Theories*

- **Example:** *Polynomial theories.* For a given lie group G , let $\mathbb{R}[\mathfrak{g}]^k$ the set of degree k homogeneous functions. So, for a given $\omega \in \Omega^l(M; \mathfrak{g})$ and $f \in \mathbb{R}[\mathfrak{g}]^k$, we can define a lk -form with values on $\mathbb{R}$

$$f(\omega)_p(v_1, \dots, v_{kl}) = \frac{1}{(kl)!} \sum \varepsilon_\sigma f(\omega_p(v_{\sigma(1)}, \dots, v_{\sigma(l)}), \dots, \omega_p(v_{\sigma(kl-l+1)}, \dots, v_{\sigma(kl)})).$$

So, if the manifold is kl -dimensional, then we can define an action

$$S : \Omega^l(M; \mathfrak{g}) \times \mathfrak{g}^k[t] \rightarrow \mathbb{R} \quad \text{by} \quad S(\omega, f) = \int_M f(\omega).$$

The interesting situations are ones such that $f(\omega)$ is a closed form, of mode that it defines a cohomology class on $H_{dR}^{lk}(M)$ and, therefore, the theory compute *only topological framework*. This happens when ω and f are invariant by adjoint representation $\text{ad}_g : \mathfrak{g} \rightarrow \mathfrak{g}$. In such cases, $f(\omega)$ is generally identified with characteristic classes.

- **Example:** *Yang-Mills theories.* We have theories describing forces. Indeed, let G be a Lie group and consider the set $\mathcal{A}$ of connections of G -principal bundles over a certain target space M , which we suppose endowed with a fixed metric g . Each of this can be regarded as a $\mathfrak{g}$ -valued 1-form. Such connections are like the potential for the force, which is the curvature $\mathfrak{g}$ -valued 2-form Ω . So, we expected that the action must be build using Ω . There are two ways to do this:

- if the manifold has dimension $2k$, choosing a polynomial $f \in \mathfrak{g}^k[t]$ and putting simply $S_{TYM}(A) = \int_M f(\Omega)$. So, when f is G -invariant, then the action is sensible only to the topology. We call them of *topological Yang-Mills theory defined by f* ;
- for an arbitrary dimension, remember the Hodge star operator $*_g$, which take a k form ω and return a $n - k$ form $*_g\omega$. Thus $\Omega \wedge *_g\Omega$ is an n -form with values in $\mathfrak{g}$. So, for a given map $f \in \mathbb{R}[\mathfrak{g}]^1$, we have

$$S_{YM} : \mathcal{A} \rightarrow \mathbb{R}, \quad \text{as being} \quad S_{YM}[A] = \int_M f(\Omega \wedge *_g\Omega).$$

It are called the *Yang-Mills theory defined by f* . In general, when $\mathfrak{g}$ is a matrix lie algebra, the take linear map f being simply the trace. Now, we observe that any connections A_i and A'_j that are compatible on the intersections $U_i \cap U_j$ have the same curvature and, therefore, defines the same actions (we that such connections are related by *gauge transformations*). Thus, generally S pass to quotient, being defined on the quotient space $\mathcal{A}/\mathcal{G}$ by gauge transformations.

- particular case: electromagnetism. In this case, the interaction term in just the holonomy of the connection.
- **Example:** *Chern-Simons theories.* For odd dimensions, there is another way to build gauge theories with are not exactly Yang-Mills but are related with them. Indeed, let G be a Lie group and let M be a manifold of dimension 1 endowed with a metric. To build a gauge theory, we simply put

$$S_{CS} : \mathcal{A} \rightarrow \mathbb{R} \quad \text{as being} \quad S_{CS}[A] = \int_M h(A),$$

where $h : \mathfrak{g} \rightarrow \mathbb{R}$ is any invariant linear map. In dimension three we can also define

$$S_{CS}[A] = \int_M h(A \wedge dA + aA \wedge A \wedge A).$$

In dimension five, we can put

$$S_{CS}[A] = \int_M h(A \wedge dA \wedge dA + aA \wedge dA \wedge A \wedge A + bA \wedge A \wedge A \wedge A \wedge A).$$

In dimension seven or more, the definition is analogous. They are called *Chern-Simons theories* and are closely related with the polynomial theories defined previously. Indeed, in many situations we can show that, by a given invariant polynomial function $f \in \mathbb{R}[\mathfrak{g}]^k$, the form $f(\Omega) \in \Omega^{2k}(M)$ is exact. So, there is a $2k - 1$ form $\text{cs}(A)^k$ such that $d\text{cs}(A)^k = f(\Omega)$. Happens that the previous forms

$$h(A), \quad h(A \wedge dA + aA \wedge A \wedge A), \quad \text{etc}$$

corresponds exactly to $\text{cs}(A)^0$, $\text{cs}(A)^1$, etc for certain f .

- For the case of manifolds with boundary, we can then write, by the Stokes theorem:

$$\int_M f(\Omega) = \int_M d\text{cs}(A) = \int_{\partial M} \text{cs}(A),$$

of mode that *Chern-Simons theories computes essentially the boundary part of what was called topological Yang-Mills theories*. This is an example of manifestation of the holographic principle.

- **Example: Gravity Theories.** The classical way to study gravity is looking for semi-riemannian metrics g (that is, by the $O(r, s)$ -reductions) on a fixed oriented ambient manifold M . The action functional is the *Einstein-Hilbert action*

$$S : \text{Met}(M)/\text{Diff}(M) \rightarrow \mathbb{R}, \quad \text{given by} \quad S_{EH}[g] = \int_M S_g \cdot \omega_g,$$

where S_g is the scalar curvature of g . But remember that gravity is equivalently the study of Cartan connections for the reduction $O(r, s) \hookrightarrow \text{Iso}(r, s)$. So, we expected that the Einstein-Hilbert action can also be view as an action over such Cartan connections. Indeed, let e and ω the vielbein and the spin connections corresponding to a given metric g . Then ω induces a linear connection on TM , which is a linear derivation

$$D : \Gamma(TM) \rightarrow \Gamma(TM^* \otimes TM).$$

Locally, for any fixed basis of vector fields e_i , write $De_i = \omega_i^j \otimes e_j$. When the basis is induced by the vielbein, we have $\omega_i^j = \Gamma_{ik}^j \cdot e^k$. So, the curvature Ω of D , which is a 2-form which values in TM , can be computed in the basis e^i (which we supposed provided by a chart) by

$$\begin{aligned} \Omega_i^j &= d\omega_i^j + \omega_i^k \wedge \omega_k^j \\ &= d(\Gamma_{ik}^j \cdot e^k) + (\Gamma_{il}^k \cdot e^l) \wedge (\Gamma_{km}^j \cdot e^m) \\ &= \partial_l \Gamma_{ik}^j \cdot e^l \wedge e^k + (\Gamma_{il}^k \Gamma_{km}^j) \cdot (e^l \wedge e^m) \\ &= (\partial_l \Gamma_{im}^j + \Gamma_{il}^k \Gamma_{km}^j) \cdot e^l \wedge e^m \\ &= \frac{1}{2} R_{ilm}^j \cdot e^l \wedge e^m, \end{aligned}$$

of mode that

$$\Omega^{kj} = g^{ki} \Omega_i^j = \frac{1}{2} g^{ki} R_{ilm}^j \cdot e^l \wedge e^m.$$

Now, in *dimension four* we have the 4-form

$$\begin{aligned} \Omega^{kj} \wedge *(e^k \wedge e^j) &= (g^{ki} R_{ilm}^j) \cdot (e^l \wedge e^m) \wedge *(e^k \wedge e^j) \\ &= \frac{1}{2} g^{ki} R_{ilm}^j (\delta_j^l \delta_k^m - \delta_k^l \delta_j^m) \cdot \omega_e \\ &= g^{ij} R_{ij} \cdot \omega_g. \end{aligned}$$

Therefore, if $\mathcal{A}_C$ is the collection of Cartan connections for $O(3, 1) \hookrightarrow \text{Iso}(3, 1)$, then the Einstein-Hilbert action can be identified with the *Palatini action functional*

$$S : \mathcal{A}_C \rightarrow \mathbb{R} \quad \text{given by} \quad S[\omega, e] = \int_M \text{tr}(\Omega \wedge *(e \wedge e)).$$

– Similarly, in *dimension three*, we have that the Einstein Hilbert action is

$$\begin{aligned} S[\omega, e] &= \int_M \text{tr}(\Omega \wedge *e) \\ &= \int_M \text{tr}(d\omega * e + \omega \wedge \omega * e). \end{aligned}$$

So, for suitable choice of invariant polynomial, we have $S_{CS}(A) = S[\omega, e]$. Therefore, *three dimensional gravity is purely topological*. This is interesting because in general we can quantize Chern-Simon theories, so that the previous identification allow us to consider three dimensional Chern-Simon theory for $\mathfrak{iso}(2, 1)$ as being a model to quantum gravity.

- **Example:** *Supergravity Theories.*

– The same holds for strings? We can see that

- **Example:** *WZW-theories.* Let G be a Lie group. We can consider special theories of strings moving inside G . Let $h \in \mathbb{R}[\mathfrak{g}]^2$ be some invariant map which we supposed non-degenerated. So, for a given two dimensional manifold Σ , we can consider the Polyakov action for f . That is, we can consider

$$S_{Pol}[\varphi, g] = \int_{\Sigma} h(\partial_i \varphi, g_{ij} \partial_j \varphi) \sqrt{g} \cdot dx^1 \dots dx^n.$$

Now, observe that h induces a 3-form $\omega = h(-, [-, -])$ on $\mathfrak{g}$, which extends to a 3-form on the Lie group G that also will be denoted by ω . This allow us to consider another interaction term. Indeed, for a given immersion $\iota : \Sigma \hookrightarrow M$, where M is

a three manifold, suppose that any φ admits an extension $\tilde{\varphi}$ to M . Then $\tilde{\varphi}^*\omega$ is a volume form on M , and then we can consider the term

$$S_{WZW}[\varphi] = \int_M \tilde{\varphi}^*\omega.$$

.....existence and uniqueness of extensions.

- The interest in the WZW model is by the following: differently of Yang-Mills theory, Chern-Simons theory defined in a manifold with boundary is not invariant by gauge transformations. Furthermore, the failure of invariance is given by a term $\int_{\partial M}$. Such term is exactly the WZW term for.
- some problems which generalizations of the previous example and solutions given by passing from higher geometry.

2.3 Motion

- Now, remember that, as was said in the previous chapter, *classical theories* are just ones that select the possible
- **Example:** *Kinetic Actions:* Geodesics.
- **Example:** *Spinors Theories.* Dirac equation.
- **Example:** *Gauge Theories.* Yang-Mills equations. Maxwell equations
- **Example:** *Chern-Simons.* Flat connections.
- **Example:** *Gravity.* Einstein equation.

Boundaries, D-Branes, Black Branes and Defects

- instantons
- solitons
- black holes,
- etc.

Overview

- definition of classical theory

Chapter 3

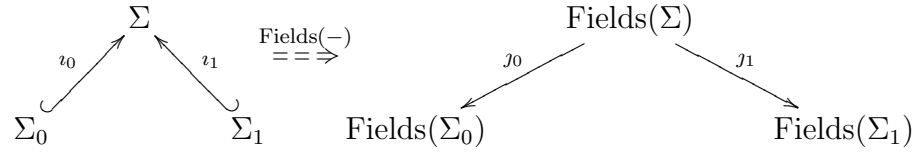
Abstraction

- in the previous chapter was briefly discussed the role of geometry in the study of classical physics. Here we will discuss the (higher) categorical theoretical aspects of those theories.
- in the first two sections we will see explicitly that a more abstract language provides more richer tools, *simplifying the analysis* of many problems and *clarifying the real nature* of many objects. More precisely, in the first section we will see that in a sufficiently abstract context, any interesting classical theory can be considered a sigma-model theory. This means that abstract languages allow us to simplify the analysis of some things. On the other hand, in the second section we will see that such generalized sigma-models are, indeed, certain cohomology theories, so that the addition of tools really clarify the nature of many things.
- third and fourth sections: synthesis.... we will see what we really need to develop classical theory.
- In comparison to the previous chapters, the following is much more abstract and, at the same time, a little imprecise by the mathematical viewpoint. On the other hand, the discussion here present is very motivating for the concepts and results that will be present in the last two parts of the texts. So, we would like to

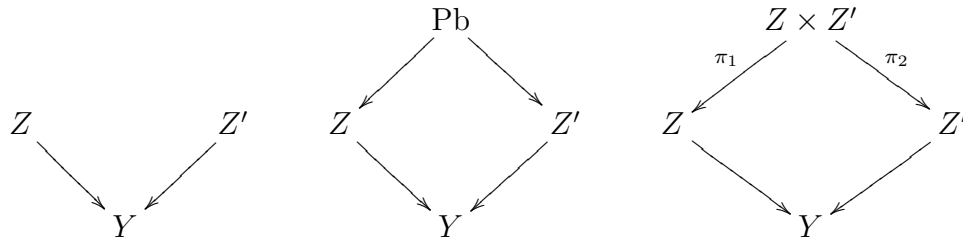
3.1 Fields

The most simple examples of phase spaces are those given by the space of maps between two manifolds, i.e $\text{Fields}(\Sigma) = C^\infty(\Sigma; M)$, which corresponds to the sigma model theories. These are very well behaved. For example, we can talk about the “initial phase space” $\text{Fields}(\Sigma_0)$ and the “final phase space” $\text{Fields}(\Sigma_1)$, where here we are considering the manifolds Σ_0 and Σ_1 as cobordisms from itself to $\emptyset$. So, the rule Fields maps any object $\Sigma_0 \in \mathbf{Cob}_p$ into a set as well as any cobordism $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ into a diagram in $\mathbf{Set}$,

as present below.



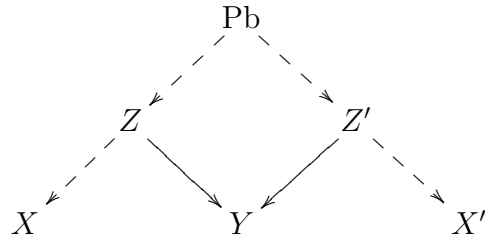
A diagram like the second is called a *span* in **Set**. So, a span of sets is simply given by two functions having the same domain. Analogously, we can talk about *cospan*s of sets, that are functions have the same codomain. We know how to “complete” any cospan of sets in an universal way. More precisely, given a cospan as in the first below diagram there is a set Pb as well as maps $Pb \rightarrow Z$ and $Pb \rightarrow Z'$ (that is, a span) such that the second diagram is commutative and universal. Indeed, Pb is simply the biggest subset of $Z \times Z'$ in which the third diagram commutes and the required maps are the restrictions of the projections to them (the completion of cospan corresponding to the process of computing a *pullback between two maps*, what will be discussed with details in the section ??).



Now, thanks to the possibility to complete any spans of sets, we also know how to compose any two of them. Indeed, given



its composition is the following span of dotted arrows, where the top dotted arrows are the universal completion of the continuum arrows.



So, we have a category of spans of sets, which will be denoted by $\text{Span}(\mathbf{Set})$. In **Set** we also have the notion of universal completion of spans, so that we can compose cospan,

forming a category of cospans. Observe that cobordisms are a certain kind of cospan of manifolds and the composition of cobordisms is obtained (as sets) by the composition the underlying cospans. Therefore, the rule **Fields** maps manifolds into sets, as well as cobordisms into spans of sets in such way that compositions are preserved. This means that the phase space of sigma-model theories extends to a functor

$$\mathbf{Fields} : \mathbf{Cob}_p^{op} \rightarrow \mathbf{Span}(\mathbf{Set}).$$

There is a natural monoidal structure in the category of sets. Such structure induces a new monoidal structure in $\mathbf{Span}(\mathbf{Set})$, whose product $\otimes$ is obtained as below. We can verify that the previous phase space functor preserves such monoidal structure, which will be very important in the step of (pull-push) quantization, discussed in the section (5.3). So, the sigma-model theories really are very well behaved.

*****DIAGRAM*****

But a theory defined on $C^\infty(\Sigma; M)$ describes only *bosonic branes free of external interactions*, which is a very restrictive case. So we can ask if there exists other theories describing more complex situations which also are well behaved. More precisely, given a certain phase space $\mathbf{Fields}(\Sigma)$, we can ask if there is a monoidal category with pullbacks $\mathbf{H}$, here called a *context*, in which it extends to a monoidal functor

$$\mathbf{Fields} : \mathbf{Cob}_p \rightarrow \mathbf{Span}(\mathbf{H}).$$

By the previous discussion, this is immediately obtained if the category $\mathbf{H}$ have internal hom-objects and if there is some $\mathbf{Fields} \in \mathbf{H}$, called *classifying space* for the phase space functor, such that

$$\mathbf{Fields}(\Sigma_0) \simeq [\Pi(\Sigma_0); \mathbf{Fields}]_{\mathbf{H}}, \quad \text{for any } \Sigma_0, \quad (3.1)$$

where here Π is some well behaved functor from the category of manifolds to $\mathbf{H}$. Observe that a theory satisfying such conditions is something like *a sigma-model internal to the context $\mathbf{H}$* . So, being more specific: *can we see general theories as being sigma-model theories internal to a sufficiently abstract ambient space?*

As will be explicitly discussed in the next subsections, any of the standard examples of classical theories really can be seen as a sigma-model internal to some context! We recall that these examples involves the following quantities or even some combination (say products) between them:

- the set of *smooth maps* $C^\infty(\Sigma; M)$, for bosonic theories;
- the set of *sections* $\Gamma(E)$ of some fixed bundle $E \rightarrow \Sigma$, for fermionic theories;
- the set of *supermaps* $C^\infty(\Sigma; M)$, for supersymmetric theories;
- the set of *connections* over some G -bundle over Σ , for Yang-Mills as well as for Chern-Simons theories;

- the set of G -structures for classical gravity.

Furthermore, in some cases we need to consider symmetries (or even higher symmetries), so that the phase spaces are generally not only sets, but groupoids (or even ∞ -groupoids). Therefore, our work will be to build

1. monoidal categories $\mathbf{H}$ with pullbacks and internal hom and that contains the categories $\mathbf{Diff}$ and $\mathbf{Gpd}_\infty$;
2. such that the rules $\Sigma_0 \mapsto \mathbf{Fields}(\Sigma_0)$, where $\mathbf{Fields}(\Sigma_0)$ is any of the above quantities (say smooth maps, sections, supermaps, connections, etc.) are well defined and can be extended to functors $\mathbf{Fields} : \mathbf{Cob}_p \rightarrow \mathbf{Span}(\mathbf{H})$;
3. for which there exists some classifying space $\mathbf{Fields} \in \mathbf{H}$.

As we will see, in any of the above cases (say for smooth maps, sections, etc.) the main idea is to consider $\mathbf{H}$ as being a category derived from certain special functors $\mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$, called ∞ -stacks, and $\mathbf{Fields}$ as being the ∞ -stack that naturally classify the structure in analysis. In the following subsections we will motivate the introduction of ∞ -stacks and in the sequence we will analyse separated each of the previous cases, showing that they really can be seen as sigma-models.

Strategy

The first above condition requires that $\mathbf{Diff}$ (as well as $\mathbf{Gpd}$ and $\mathbf{Gpd}_\infty$) be embed in some nice category $\mathbf{H}$. So, our starting point is to get categories in which such embeds exists. There is a result, the *Yoneda embedding* (that will be discussed in the section ??), saying that any category $\mathbf{C}$ (particularly $\mathbf{Diff}$) can be embed in the category of functors $\mathbf{Func}(\mathbf{C}^{op}; \mathbf{Set})$. Such category is very well behaved. For instance, it satisfy almost all the requirements of the first condition: it is monoidal, with product given by

$$(F \otimes G)(X) = F(X) \times G(X),$$

it has pullbacks (also defined componentwise) as well as internal hom given

$$[F, G](X) = \mathbf{Mor}_{\mathbf{Func}(\mathbf{C}^{op}; \mathbf{Set})}(F \otimes X; G) = \mathbf{Nat}(F \otimes X; G),$$

where we are regarding X as a functor. So, in order to consider such category as being the context $\mathbf{H}$, we only need to get an embed $\mathbf{Gpd}_\infty \hookrightarrow \mathbf{Func}(\mathbf{C}^{op}; \mathbf{Set})$. But the first is an ∞ -category while the second is an usual category, so that any of such embed will necessarily forget higher categorical information, which is not a good thing. So, the idea is to replace the functor category by the higher categorical version $\mathbf{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty)$, where here we are regarding $\mathbf{Diff}$ as being a ∞ -category. By induction we can see that the properties satisfied by the functor category also is satisfied by its higher version with an unique

modification:..... commutativity must be replace by commutativity up to higher morphisms.....

Observe that as present below we have two natural functors, defined by $\Gamma(F) = F(*)$ and by $\text{Cst}(S)(X) = S$, where $*$ is the 0-manifold with an unique element. So, we can use the constant functor Cst to embed any ∞ -groupoid into $\text{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty)$, motivating us to consider it as being the context $\mathbf{H}$ internal to which the classical theories will be seen as sigma-models.

$$\text{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty) \begin{matrix} \xrightarrow{\Gamma} \\ \xleftarrow{\text{Cst}} \end{matrix} \mathbf{Gpd}_\infty$$

Now, for each particular situation (smooth maps, sections, etc.) we need to analyse if the other two conditions is satisfied for such choice of context. More precisely, for any of such situation we need to show that the phase space extends to a functor

$$\text{Fields} : \mathbf{Cob}_p \rightarrow \text{Span}(\mathbf{H})$$

which is totally determined by a special element $\mathbf{Fields} \in \mathbf{H}$, in the sense that (3.1) is satisfied. In order to do this, observe that the phase space rules, say for instance $C^\infty(-, M)$, generally extends to functors $F : \mathbf{Diff}^{op} \rightarrow \infty\mathbf{Gpd}$ and therefore can be seen as objects of $\text{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty)$. So, having done such extension, the main idea is to search, in each case, for a certain kind of “universal structure” that classify the associated $F(\Sigma_0)$ for any Σ_0 , so that we can take $\mathbf{Fields}$ as being such universal structure. Finally, defining

$$\text{Fields}(-) : \mathbf{Cob}_p^{op} \rightarrow \text{Span}(\mathbf{H}) \quad \text{by} \quad \text{Fields}(\Sigma_0) = [\Sigma_0; \mathbf{Fields}]$$

at objects and by the below diagram at morphisms we see that (3.1) is immediately satisfied. Therefore, the main problem is to extend the phase space rules to functors $\mathbf{Diff}^{op} \rightarrow \infty\mathbf{Gpd}$ and then to get an universal functor that classify the phase space.

***** DIAGRAM*****

Now, let is say that such classification part can be effectively done in *any* case. More precisely, we have a result, called Yoneda lemma (from which we get the Yoneda embedding) stating that for any functor $F : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ and any manifold X we have an equivalence of ∞ -groupoids

$$F(X) \simeq \text{Mor}_{\text{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty)}(X, F) = \text{Nat}_\infty(X, F),$$

so that $F(X)$ is actually classified by F when we regard manifolds as ∞ -functors. Therefore, *in order to see an arbitrary classical theory as being a sigma-model internal to $\text{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty)$, we only need to extend its phase space to an object of such context, i.e to a functor $\mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$.*

Locality

Before to analyse each situation separately, there is another thing that can be said. Indeed, all involved structures (say smoothness of maps, sections and connections of bundles) are *local*. This means that they are totally determined along any open cover. More precisely, if $F : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ is some ∞ -functor representing one of those phase space functors, then for a given manifold $X \in \mathbf{Diff}$, the choice of cover $U_i \hookrightarrow X$ determines any $f \in F(X)$ uniquely in terms of a family (f_i) , with $f_i \in F(U_i)$, such that:

1. is compatible along the overlaps $U_{ij} = U_i \cap U_j$;
2. as well as along $U_{ijk} = U_i \cap U_j \cap U_k$;
3. and along $U_{ijkl} = U_i \cap U_j \cap U_k \cap U_l$;
4. and so on.

In explicit terms, these “compatibility conditions” can be described in the following way: observe that we have a sequence of mappings, as present below. Any of such maps are universal inclusions between manifolds and, therefore, can be regarded as mappings between ∞ -functors. Its limit (here called a colimit, because the sequence decreases) is denoted by $\check{C}(U_i) = \text{colim} U_{ijk} \dots$ and called the *Cech nerve* of the cover. It comes equipped with a canonical map $\check{C}(U_i) \rightarrow X$: the *Cech cover* of X associate to $U_i \hookrightarrow X$.

*****diagram*****

We observe that the functor F induces the below dual sequence. Any of such dual maps are called *restrictions*, being denoted by e.g by $(-)|_{U_{ij}}$. Recall that F takes values in $\mathbf{Gpd}_\infty$, so that any of the above terms are ∞ -groupoids and, therefore, have objects, morphisms, 2-morphisms, and so on.

***** diagram*****

Now we can describe precisely the compatibility conditions over the family (f_i) . Indeed, it means simply that there is

1. for any i and j there is a morphism $f_{ij} : f_i|_{U_{ij}} \rightarrow f_j|_{U_{ij}}$;
2. for any i, j and k there is a 2-morphism f_{ikj} between $f_{ij}|_{U_{ijk}}$ and $f_{ik}|_{U_{ijk}} \circ f_{kj}|_{U_{ijk}}$;
3. and so on.

Observe that any f_i is an object of $F(U_i)$. So, what we are saying is that any $f \in F(X)$ is totally determined by the following data:

1. for any i , an object $f_i \in F(U_i)$;
2. for any i and j , a morphism $f_{ij} : f_i|_{U_{ij}} \rightarrow f_j|_{U_{ij}}$;
3. for any i, j and k , a 2-morphism $f_{ikj} : f_{ij}|_{U_{ijk}} \Rightarrow f_{ik}|_{U_{ijk}} \circ f_{kj}|_{U_{ijk}}$;

4. and so on.

The fact that any $f \in F(X)$ is *totally* described by the previous data also can be put in precise terms. Taking the “limit” of the above diagram of ∞ -groupoids we get a new ∞ -groupoid $\lim F(U_{ijk\dots})$ together a canonical mapping $\lim F(U_{ijk\dots}) \rightarrow F(X)$. Saying that any f is uniquely described by (f_i) means precisely that the canonical mapping is an equivalence.

A ∞ -functor $F : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ such that for any X the ∞ -groupoid $F(X)$ is totally determined along any cover $U_i \hookrightarrow X$ as described above is called a *smooth stack of ∞ -groupoids* (or simply a *smooth ∞ -stack*). So, what we are saying is that in general the phase space of classical theories extends not only to ∞ -functors, but (thanks to its local properties) to smooth ∞ -stacks. So, the natural context $\mathbf{H}$ internal to which classical theories always can be seen as sigma-models do not is exactly $\mathbf{Func}_\infty(\mathbf{Diff}^{op}; \mathbf{Gpd}_\infty)$, but the subcategory $\mathbf{Smooth}\infty\mathbf{Stack}$ of smooth ∞ -stacks. Because of this, contravariant ∞ -functors are also generally called *smooth ∞ -pre-stacks*.

Such subcategory have very suitable properties, turning it what is called an *∞ -topos* (such entities will be discussed in the chapter ??). For example, this means that Γ and $\mathbf{Cst}$ have adjoints.....

It is further more special: it have an additional structure of *cohesion*, making it a *cohesive ∞ -topos*..... Such cohesion is what makes possible talking about differential geometry inside it.

Presentation

Observe that the notion of smooth ∞ -stack depends explicitly of the notions of ∞ -groupoids and ∞ -functors. These notions, on their turn, can be formalized (or present) in many different ways. So, different presentations for such higher categorical notions will produce different formalization for the concept of smooth ∞ -stacks. Here we will comment very briefly on how to give such presentations for stacks. A more detailed discussion will be given in the chapter ??.

We initiate recalling that, as was commented at the section ?? (and as will be discussed in the section ??), the higher categorical notions are generally formalized in terms of *simplicial sets*. These simplicial sets, on the other hand, can be obtained from usual 1-categories $\mathcal{C}$ with a distinguished and suitable class of morphism $W \subset \mathcal{C}$, called *weak equivalences*. Using such data we can build a new category $\mathcal{C}[W^{-1}]$, called the *localization* of $\mathcal{C}$ at W , in which any $f \in W$ becomes an isomorphism (the process is very similar to the localization of a ring at a distinguished part of a ring).

Now, the main idea is to consider $\mathbf{Smooth}\infty\mathbf{Stack}$ as being a certain kind of localization of the category of ∞ -functors $\mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ at some special class of morphisms. So, we need first identify a presentation of such functor category and then choose a suitable class of morphisms to be localized in order to produce the data described by smooth ∞ -stacks. We observe that presenting $\mathbf{Gpd}_\infty$ by simplicial sets, it is expect that ∞ -functors

$\mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ be presented by usual functors $\mathbf{Diff}^{op} \rightarrow s\mathbf{Set}$. This is indeed the case, as will be discussed in the section ??.

In order to identify the morphisms to be localized, observe that if $U_i \hookrightarrow X$ is a cover of a manifold and F is some ∞ -functor, then the canonical map $F(X) \rightarrow F(U_{ijk\dots})$ (which need to be an equivalence if we want F to be a smooth ∞ -stack) can be obtained from $\check{C}(U_i) \rightarrow X$ in the following way

$$F(X) \simeq \mathrm{Nat}_\infty(X; F) \rightarrow \mathrm{Nat}_\infty(\mathrm{colim} U_{ijk\dots}, F) \simeq \lim \mathrm{Nat}_\infty(U_{ijk}, F) \simeq \lim F(U_{ijk\dots}),$$

where in the first step was used the Yoneda lemma, in the second was applied $\mathrm{Nat}_\infty(-, F)$ to the morphism $\check{C}(U_i) \rightarrow X$, in the third was used the fact that hom-functors maps colimits into limits and, finally, the Yoneda lemma was applied again. Therefore,

$$\check{C}(U_i) \simeq X \quad \text{imply} \quad F(X) \simeq \lim F(U_{ijk\dots})$$

for any F . This indicate that localizing the category of ∞ -functors at the canonical morphisms between Cech covers and the underlying manifold we get precisely the category of smooth ∞ -stacks. So, being ∞ -functors $\mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ presented by functors $\mathbf{Diff}^{op} \rightarrow s\mathbf{Set}$ is expect that $\mathrm{Smooth}\infty\mathrm{Stack}$ be presented by

$$\mathrm{Func}(\mathbf{Diff}^{op}; s\mathbf{Set})[W^{-1}], \quad \text{with} \quad W = \{\check{C}(U_i) \rightarrow X\},$$

as will be confirmed in the section ??. Such discussion can be summarized in this table:

higher structures	simplicial presentation
∞ -categories	simplicial sets
∞ -functors	simplicial functors
smooth ∞ -stacks	simplicial functors localized at Cech covers

Table 3.1: Higher entities and their simplicial presentations

Geometric Stacks

- any representable ∞ -functor is a smooth stack
- Lie groupoids
- in the ambient category $\mathbf{Diff}$ the

Cohomology

- flavors of cohomology:
 - hom-groupoids, corresponding to non-abelian cohomologies;
 - stabilization.

3.2 Assertion

In this section we will see that the standard classical theories really can be considered as sigma-models internal to the category smooth ∞ -stacks (or to some other nice category directly obtained from it).

We initiate considering the case in which the phase space (or some part of it) is given by the set of smooth maps $C^\infty(\Sigma, M)$ into a fixed ambient space M (that is, the classical sigma-model theories). It appears typically for theories describing bosonic p -branes and can be trivially extended to a functor $C_M^\infty : \mathbf{Diff}^{op} \rightarrow \mathbf{Set}$ (therefore, thanks to the inclusion $\mathbf{Set} \hookrightarrow \mathbf{Gpd}_\infty$, to a ∞ -functor from the category of manifolds to the category of ∞ -groupoids).

By the Yoneda lemma we have $C_M^\infty(X) \simeq \mathrm{Nat}_\infty(X; C_M^\infty)$ so that to give a smooth map $X \rightarrow M$ is just the same that giving a morphism $X \rightarrow C_M^\infty$. Therefore, C_M^∞ classify the phase space in analysis and can be taken as being the universal functor **Fields**. We observe that C_M^∞ is exactly the ∞ -functor defined by the manifold M regarded as a Lie groupoid and, therefore, really is a smooth stack. Therefore, *the classical sigma-model theories can be seen as a sigma-model internal to the category of smooth ∞ -stacks with classifying space given by C_M^∞ .*

In order to clarify the ideas, let see explicitly that C_M^∞ is a stack. So, given a cover $U_i \hookrightarrow X$ we need to show that

$$\lim C_M^\infty(\check{C}(U_i)) \simeq C_M^\infty(X).$$

Observe that, because C_M^∞ initially takes values in **Set**, all the ∞ -groupoids $C_M^\infty(U_{ij\dots})$ have trivial morphisms, trivial 2-morphisms and so on. Thus, the data contained on the sequence $C_M^\infty(\check{C}(U_i))$ is

1. for any i , an element $f_i \in C_M^\infty(U_i)$, i.e an smooth map $U_i \rightarrow M$;
2. for any i and j , the trivial morphism $id : f_i|_{U_{ij}} \rightarrow f_j|_{U_{ij}}$;
3. for any i, j and k the trivial 2-morphism $id : f_{ij}|_{U_{ijk}} \Rightarrow f_{ik}|_{U_{ijk}} \circ f_{kj}|_{U_{ijk}}$;
4. and so on.

Consequently, the condition $\lim C_M^\infty(\check{C}(U_i)) \simeq C_M^\infty(X)$ means that, chosen $U_i \hookrightarrow X$, any smooth map $f : X \rightarrow M$ is uniquely determined by a family of smooth maps $f_i : U_i \rightarrow M$ such that $f_i|_{U_{ij}} = f_j|_{U_{ij}}$, which is clearly true.

Smooth Supermaps

- concept of ∞ -stack depends only of the notions of cover and intersections;
- site of cartesian spaces. Equivalence between smooth stacks and cartesian stacks;

- site of supermanifolds;
- comment about stacks on other sites and taking values in other categories and their importance in algebraic geometry and in physics (for instance, the

Smooth Bundles

Here we will consider the classical theories whose phase space is the set $\text{Iso}_G(X)$ of smooth G -bundles (up to isomorphisms) over a fixed space X . Observe that because we are considering G -bundles up to isomorphisms they defines a groupoid $\text{Princ}_G(X)$. So, our phase space extends to a functor $\text{Princ}_G : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}$ and, therefore, to a functor taking values in $\mathbf{Gpd}_\infty$. By the Yoneda lemma, we have

$$\text{Princ}_G(X) \simeq \text{Nat}_\infty(X, \text{Princ}_G),$$

so that we can effectively classify all G -principal bundles over X regarding manifolds as being ∞ -functors. We assert that such functor is indeed a smooth stack. More precisely, the Lie group G induces a natural Lie groupoid $\mathbf{BG}$: its *delooped groupoid*, whose set of objects is the trivial manifold $*$ and whose set of morphisms is G . We will prove that, for any $U_i \hookrightarrow X$ cover of any manifold X , the groupoid $C^\infty(C(U_i); \mathbf{BG})$ of smooth functors and smooth transformations between them is precisely $\text{Princ}_G(X)$.

Consequently, Princ_G is just the functor induced by $\mathbf{BG}$ and, therefore, really is a smooth stack. This means that *theories whose phase space is the set of G -bundles up to isomorphisms can be seen as sigma-models internal to the category of smooth ∞ -stack, having $\mathbf{BG}$ (or equivalently Princ_G) as being the classifying space.*

Observe that a smooth functor $F : C(U_i) \rightarrow \mathbf{BG}$ in objects is the constant function $\coprod U_i \rightarrow *$ and at morphisms is determined by a family of smooth functions $g_{ij} : U_i \cap U_j \rightarrow G$ preserving the composition and the identities, meaning that

$$g_{ij} \cdot g_{jk} = g_{ik} \quad \text{and} \quad g_{ii} = 1. \quad (3.2)$$

On the other hand, a natural transformation $\xi : F \Rightarrow F'$ between two of such smooth functors is determined by a family of smooth functions $t_i : U_i \rightarrow U'_i$ such that $g'_{ij} \cdot t_i = t_j \cdot g_{ij}$. But by the local classification theorem of G -bundles, the data (3.2) corresponds precisely to a such smooth bundle and two bundles are equivalent exactly when the datas are linked by the relation $g'_{ij} \cdot t_i = t_j \cdot g_{ij}$, so that we really have

$$\text{Princ}_G(X) \simeq C^\infty(C(U_i); \mathbf{BG}).$$

For instance, in such context of Lie groupoids, the G -bundle providing from the smooth functor F is obtained considering the higher pullback (that is, the universal cospan), usually called the *homotopy fiber* of F , as present in the first below diagram. Such universal higher pullback can be computed replacing $* \rightarrow \mathbf{BG}$ by a suitable morphism $G//G \rightarrow \mathbf{BG}$, obtained looking $*$ as being the quotient of G by itself. As will be discussed in the chapter

??, this kind of replacement is a standard step in the calculation of higher limits when the underlying category is present by certain category with weak equivalences, as happens with the category of Lie groupoids.

*****DIAGRAM*****

Now, recall that a G -principal bundle is a smooth map $P \rightarrow X$ that comes endowed with an action $G \times P \rightarrow P$ in such way that $X \simeq P/G$. So, in order to proof that the pullback by F really is a G -bundle we need to get such action. It can be obtained directly from the universality of the pullbacks, as in the third diagram.

Connections

Let us consider the cases of theories having phase space given by the set of connections of a fixed G -principal bundle $P \rightarrow X$. Generally such connections are considered only up to gauge transformations. Important examples happens in Yang-Mills and Chern-Simons theories. So, in such cases we have the groupoid $\text{Conn}_G(P)$ of connections on P . In order to see these theories as sigma-models, the first step is extend the phase space to a functor $\text{Conn}_G : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$. Observe that this cannot be done immediately, because we need to consider the *whole bundle* $P \rightarrow X$ and *not only the base manifold* X . But, by the previous example, such bundle is present by a map $X \rightarrow \mathbf{BG}$ into a *fixed* stack $\mathbf{BG}$.

If $\mathbf{C}$ is an arbitrary category and $A \in \mathbf{C}$ is a *fixed* object we can always build the *over category* $\mathbf{C}/A$ whose objects are maps $X \rightarrow A$ and whose morphisms between $X \rightarrow A$ and $Y \rightarrow A$ are maps $X \rightarrow Y$ such that the first below diagram is commutative (for details, see the section ??). More generally, if $\mathbf{C}$ is a higher category, recall that we can talk about weak commutative diagrams, so that is more natural to define the morphisms of $\mathbf{C}/A$ as being given by maps $X \rightarrow Y$ together a homotopy commutative triangle, as in the second diagram.

Example 3.2.1. *****DIAGRAM

So, the main idea is to look to the slice context $\text{Sm}\infty\text{Gpd}/\mathbf{BG}$. In it each connection ∇ would be characterized by a map between $X \rightarrow \mathbf{BG}$ and some another fixed morphism of smooth ∞ -stacks $\mathbf{BG}_{\text{conn}} \rightarrow \mathbf{BG}$, as present above. In order to identify the stack $\mathbf{BG}_{\text{conn}}$, let us consider first the case in which the background bundle P is trivial. So, the rules $g_{ij} : U_{ij} \rightarrow G$ that locally characterizes P are restrictions of a global function $g : X \rightarrow G$, as well as a connection of P is simply a global 1-form $A \in \Omega^1(X; \mathfrak{g})$ which is invariant by g , i.e such that $A = g^{-1}Ag + g^{-1}dg$. Together, they define a groupoid (and, therefore, a ∞ -groupoid) $\mathbf{BG}_{\text{conn}}(X)$, whose set of objects is $\Omega^1(X; \mathfrak{g})$ and whose morphisms $A \rightarrow A'$ are the sets of maps $g \in C^\infty(X; G)$ satisfying the gauge transformation law. The rule $\mathbf{BG}_{\text{conn}}(-)$ really is a smooth ∞ -stack as can be easily verified.

Observe that the difference between $\mathbf{BG}_{\text{conn}}(X)$ and $\mathbf{BG}(X)$ is that the second has an unique object, so that for any X there is a canonical map $\mathbf{BG}_{\text{conn}}(X) \rightarrow \mathbf{BG}(X)$ and, therefore, we have a canonical morphism between the underlying stacks. The reader can

be verify that maps between $X \rightarrow \mathbf{B}G$ and those $\mathbf{B}G_{\text{conn}} \rightarrow \mathbf{B}G$ really are equivalent to connections (up to gauge transformations) on the bundle classified by $X \rightarrow \mathbf{B}G$.

Therefore, *theories whose phase space is the set of connections on a G -bundle can be seen as sigma-models internal to the slice category $\text{Smooth}\infty\text{Stack}/\mathbf{B}G$, having classified space given by the canonical $\mathbf{B}G_{\text{conn}} \rightarrow \mathbf{B}G$.*

- The above discussion clarify an important thing: suppose that need to look certain theory as a sigma model internal to some context $\mathbf{H}$. So, *if such theory depends of a background ingredient which was previously classified in another context $\mathbf{H}'$, say with classifying space A , so the main idea is to consider $\mathbf{H}$ as being $\mathbf{H}'/A$.*

Sections

Here we will consider theories whose phase space is given by the space of sections $\Gamma(E)$ of a fixed vector bundle $E \rightarrow X$, say with typical fiber V , over a manifold X . So, we have to consider here a background information: the fixed bundle. In order to apply the previous strategy, we need to classify vector bundles over X . But recall that any of them is associated to some G -principal bundle for some action $\rho : G \times V \rightarrow V$. Principal bundles are classified on $\text{Smooth}\infty\text{Stack}$ by a map $X \rightarrow \mathbf{B}G$. So, the idea is to incorporate the action ρ in some map $\bar{\rho} : \mathbb{V} \rightarrow \mathbf{B}G$ and then identify the sections of $E \rightarrow X$ as being morphisms on the slice category $\text{Smooth}\infty\text{Stack}/\mathbf{B}G$ as present below.

Having done such identification, *theories whose phase space is $\Gamma(E)$ will be seen as sigma-models internal to the slice category of ∞ -smooth stacks by $\mathbf{B}G$, having classified space given by the canonical map $\bar{\rho}$.*

$$\begin{array}{ccc} X & \overset{s}{\dashrightarrow} & \mathbb{V} \\ & \searrow \quad \swarrow & \\ & \mathbf{B}G & \end{array}$$

In order to identify the ∞ -stack $\mathbb{V}$ as well as the canonical map $\mathbb{V} \rightarrow \mathbf{B}G$, recall that any quotient induces a natural action groupoid (and, therefore, a ∞ -groupoid) and any ∞ -groupoid can be regarded as the constant stack. But any action induces a quotient: its orbit space. So, the idea is to consider $\mathbb{V}$ as being the stack $V//\rho$ defined by the orbit space of ρ . We need to build a morphism $\bar{\rho} : V//\rho \rightarrow \mathbf{B}G$. This means that for any manifold X we need to build a natural map

$$\bar{\rho}_X : (V//\rho)(X) \rightarrow \mathbf{B}G(X)$$

of ∞ -groupoids (that is, a ∞ -functor). Recall that $V//\rho$ has the elements $v \in V$ as objects and a morphism $v \rightarrow w$ is an element $g \in G$ such that $\rho(g, v) = w$ (the other higher morphisms are trivial). On the other hand, for any X , the ∞ -groupoid $\mathbf{B}G(X)$ is that associated to the group $C^\infty(X; G)$. Therefore, it has an unique object $*$ and the morphisms $* \rightarrow *$ are just the smooth functions $X \rightarrow G$ (the other higher morphisms also

are trivial). So, we can define $\bar{\rho}$ as being constant at objects, i.e $\bar{\rho}(v) = *$ for any $v \in F$, as well as for any morphism $g : v \rightarrow w$ we take $\bar{\rho}(g) : * \rightarrow *$ as being the constant function at g . Because the action ρ preserve the multiplication and the identity of the group G we see that the rule $\bar{\rho}$ really is a map between groupoids (and, therefore, between the underlying ∞ -stacks). Such map is natural because secretly it is independent of the manifold X .

So, let us prove that to give a section $s : X \rightarrow E$ of the vector bundle $E \rightarrow X$ associated to some G -principal bundle, say classified by $X \rightarrow \mathbf{B}G$, with action $\rho : G \times V \rightarrow V$, is really the same as giving a morphism $\bar{s} : X \rightarrow V//\rho$ in the slice category $\text{Smooth}\infty\text{Stack}/\mathbf{B}G$ between $X \rightarrow \mathbf{B}G$ and $\bar{\rho}$. We start recalling that, as was discussed later, as a Lie groupoid, a principal G -bundle over X is determined by the first pullback below. A G -bundle has G as typical fiber. The associated vector bundle is obtained replacing the fiber G by V and the standard action $G \times G \rightarrow G$ by $\rho : G \times V \rightarrow V$. So, the induced bundle can be computed by the second pullback below.

$$\begin{array}{ccc} P & \longrightarrow & G//G \\ \downarrow & & \downarrow \\ X & \longrightarrow & \mathbf{B}G \end{array} \quad \begin{array}{ccc} E & \xrightarrow{a} & V//\rho \\ \pi \downarrow & & \downarrow \\ X & \longrightarrow & \mathbf{B}G \end{array}$$

Recall that a section for $\pi : E \rightarrow X$ is a map $s : X \rightarrow E$ such that $\pi \circ s = id$. Now, it is easy to see that any section induces a morphism $\pi \rightarrow \bar{\rho}$ in $\text{Smooth}\infty\text{Stack}/\mathbf{B}G$: it is the dotted arrow in the first below diagram, obtained from commutativity. Reciprocally, as present in the second diagram, given one of such morphisms, say $b : \pi \rightarrow \bar{\rho}$, by commutativity it induces the dotted arrow and by universality we get the required section.

$$\begin{array}{ccc} X & & \\ \searrow^{s} & \nearrow^{a \circ s} & \\ & E & \xrightarrow{a} V//\rho \\ \downarrow \pi & \nearrow \text{dotted} & \downarrow \\ X & \longrightarrow & \mathbf{B}G \end{array} \quad \begin{array}{ccc} X & & \\ \searrow^{s} & \nearrow^{b} & \\ & E & \xrightarrow{a} V//\rho \\ \downarrow \pi & \nearrow b & \downarrow \\ X & \longrightarrow & \mathbf{B}G \end{array}$$

- any classical theory also can be seen as the sections of a bundle. Indeed, as was discussed they corresponds to sigma-models and sigma-models corresponds to space of sections of trivial bundles.

G-Structures

Abstraction

- thanks to the previous characterization we can talk about G -bundles internal to $\mathbf{H}$, as well as G -bundles with connections internal to $\mathbf{H}$, etc.

- at this moment was only showed that the standard examples of classical theories can always be described as sigma-models internal to some abstract context. Here we will discuss that this abstract language can be used not only to describes things that works very well in the classical language, but also to avoid to problems that appear there.
- string-structures are part of a 2-bundle
- analog of electromagnetism for strings can be naturally describes as 2-connections on 2-bundles

3.3 Action

In the previous section was introduced the concept of *sigma-model internal to some context* $\mathbf{H}$. They are build considering an object $\mathbf{Fields} \in \mathbf{H}$ and then taking the monoidal functor

$$\mathbf{Fields} : \mathbf{Cob}_p \rightarrow \mathbf{Span}(\mathbf{H}), \quad \text{given by} \quad \mathbf{Fields}(\Sigma_0) = [\Sigma_0, \mathbf{Fields}].$$

Was also saw that the phase space of the standard classical theories can be considered as being sigma-models internal to the context $\mathbf{SmoothSuper}\infty\mathbf{Stack}$ of smooth super ∞ -stacks. But *a classical theory is not composed only by its phase space*: we also need to consider its *action functional*. In this section, we are looking to the concept of *internal action functional* for a given internal sigma-model. Such concept must be such that, if we consider a standard classical theory as being an internal sigma-model, then to its classical action functional corresponds naturally an internal action functional. More precisely, we are looking to some quantity that complete the following table:

	classical language	internal language
phase space	set $\mathbf{Fields}(\Sigma)$	monoidal functor $\mathbf{Fields} : \mathbf{Cob}_p \rightarrow \mathbf{Span}(\mathbf{H})$
action functional	map $\mathbf{Fields}(\Sigma) \rightarrow \mathbb{R}$	???

Table 3.2: Physical concepts in classical and internalized language

In order to get such concept, recall that, as was discussed in the last chapter, if a theory has space of fields given by a set X , then an action functional is simply a real map $X \rightarrow \mathbb{R}$. Generally, X depends of some manifold Σ and then we write $X = \mathbf{Fields}(\Sigma)$. So, generally the action is a map $S_\Sigma : \mathbf{Fields}(\Sigma) \rightarrow \mathbb{R}$. We can regard $\mathbb{R}$ as a Lie groupoid $\mathbf{B}\mathbb{R}$, so that with this identifications the action S_Σ can be seen as an element of the slice category $\mathbf{Gpd}_\infty/\mathbf{B}\mathbb{R}$, whose domain is $\mathbf{Fields}(\Sigma)$.

Now, recall that in the standard examples of classical theories, the phase space makes sense for any Σ and extends to a functor $\mathbf{Fields} : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty$ (indeed, it is exactly this extension that allow us to look the theory as being a sigma-model internal to $\mathbf{Smooth}\infty\mathbf{Stack}$). So, it is expected that the action functional S_Σ also makes sense for

any Σ and extends to a functor $S : \mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty/\mathbf{BR}$ such that the domain of S_Σ is precisely $\mathbf{Fields}(\Sigma)$. For instance, polyakov action.....

For any category $\mathbf{C}$ and any object $A \in \mathbf{C}$ there is a canonical functor $\mathbf{C}/A \rightarrow \mathbf{C}$ that assign to any map $X \rightarrow A$ its domain X . So, if we denote the canonical functor $\mathbf{Gpd}_\infty/\mathbf{BR} \rightarrow \mathbf{Gpd}_\infty$ by $\mathbb{R}$, what we are saying with the extension of S is that it defines precisely a morphism $S : \mathbf{Fields} \rightarrow \mathbb{R}$ in the slice category $\mathbf{Cat}/\mathbf{BR}$, as present in the first below diagram. In other words, S is just a lifting of $\mathbf{Fields}$.

$$\begin{array}{ccc}
 \mathbf{Diff}^{op} & \xrightarrow{S} & \mathbf{Gpd}_\infty/\mathbf{BR} \\
 \searrow \text{Fields} & & \swarrow \mathbb{R} \\
 & \mathbf{Gpd}_\infty &
 \end{array}
 \qquad
 \begin{array}{ccc}
 \mathbf{Cob}_p & \xrightarrow{S} & \mathbf{Span}(\mathbf{H}/\mathbf{BR}) \\
 \searrow \text{Fields} & & \swarrow \mathbb{R} \\
 & \mathbf{Span}(\mathbf{H}) &
 \end{array}$$

On the other hand, when the phase space extends to a functor $\mathbf{Fields}$, it also extends to a monoidal functor $\mathbf{Fields}$ from $\mathbf{Cob}_p$ to the category of spans. So, it is expected that, similarly, if the action functional extends to a functor $\mathbf{Diff}^{op} \rightarrow \mathbf{Gpd}_\infty/\mathbf{BR}$ it also extends to another functor from cobordisms to sliced spans, as in the second above diagram. As we come to discuss, this is really the case. Therefore, the table (??) can be completed in the following way:

	classical language	internal language
phase space	set $\mathbf{Fields}(\Sigma)$	monoidal functor $\mathbf{Fields} : \mathbf{Cob}_p \rightarrow \mathbf{Span}(\mathbf{H})$
action functional	map $\mathbf{Fields}(\Sigma) \rightarrow \mathbb{R}$	sliced map $\mathbf{Fields} \rightarrow \mathbb{R}$

Table 3.3: Physical concepts in classical and internalized language

Background Gauge

3.4 Covariance

	classical language	internal language
phase space	set $\mathbf{Fields}(\Sigma)$	functor $\mathbf{Fields} : \mathbf{Cob}_p \rightarrow \mathbf{Span}(\mathbf{H})$
action functional	map $S : \mathbf{Fields}(\Sigma) \rightarrow \mathbb{R}$	sliced map $\mathbf{Fields} \rightarrow \mathbb{R}$
covariant phase space	critical locus $dS = 0$	???

Table 3.4: Physical concepts in classical and internalized language

Jets

Lagrangian

- jets

Chapter 4

Quantum

As we have seen, there two kinds of physical theories: classical and quantum. In the previous chapter, was discussed briefly the role of geometry in the description of classical theories. In this chapter, we look to the quantum theories.

4.1 Theories

A *quantum theory* is essentially a way to represent the abstract movements of a certain object in terms of vector spaces and linear transformations. More precisely, an *n-dimensional topological quantum field theory* (TQFT) of type (X, E) is a symmetric monoidal functor

$$U : (\mathbf{Cob}_n^{(M,E)}, \sqcup) \rightarrow (\mathbf{Vec}_{\mathbb{C}}, \otimes)$$

from the monoidal category of *n-dimensional cobordisms with (M, E) -structure* to the monoidal category of complex vector spaces. Here, a (M, E) -structure in a manifold Σ is simply a vector bundle $E \rightarrow M$, say of rank $n \leq k$, together a continuous mapping $\varphi : \Sigma \rightarrow M$ for any Σ , such that the pullback of E by φ gives a orthogonal complement for $T\Sigma$, in the sense that

$$\varphi^* E \simeq T\Sigma \oplus \mathbb{R}^{k-n}.$$

This definition was given by Lurie in [13] and is an extension of the classical definition given by Atiyah in [??]. To interpret it, recall that, as we have seen, p -manifolds describes physical states of p -dimensional objects, as well a cobordism Σ between two of such manifolds Σ_0 and Σ_1 is interpreted as a time evolution between the underlying physical states. But the p -branes may have certain internal degrees of freedom, which are described by additional geometric structure on Σ . The p -branes are supposed to be moving in some ambient M (which generally is required to posses larger dimension than Σ) and the additional structure on Σ are generally pulled back from other additional structure in M by some configuration $\varphi : \Sigma \rightarrow M$.

Therefore, instead classical theories describes the “classical movement” of p -branes, a *p-dimensional quantum theory of type (M, E)* describes the “quantum movement” of p -

branes inside an ambient M , having internal degrees of freedom determined by E . We observe that there is a main difference between the classical description and the quantum description of such movement. Indeed, recall that *in a classical theory, we **fix one** of all such cobordism Σ* (which corresponds to some abstract time evolution from Σ_0 to Σ_1) and then we look to some action functional. On the other hand, according to the previous definition, *in a quantum theory we work simultaneously **with all** cobordisms Σ* (that is, with all time evolution).

This is in accordance with the fact that, instead classical theories are deterministic, quantum theories are presumably probabilistic. More precisely, while in classical theories we can really talk about “the true” trajectory of a particle (which relies in the previous choose of some Σ), is expected that in quantum theories we lose such notion, forcing us to work with all Σ simultaneously. Here, “is expected” means that such requirement on the definition of TQFT provides from incomplete induction on the Max Born interpretation (the Born’s law) of certain set of very reliable experiments, which lies in the historical foundations of quantum theory.

Example 4.1.1 (*G-structured QFT*). Observe that for any given p , there is a special class of p -dimensional QTF: those in which the p -branes are moving in the classifying space BG of a certain topological group G . More precisely, are of type (BG, E_ρ) , where E is the rank p associated vector bundle of the universal bundle $EG \rightarrow BG$, defined by some representation $\rho : G \rightarrow GL(p)$. In such case, for any Σ , the homotopy classes of continuous mappings $\varphi : \Sigma \rightarrow BG$ are in bijective correspondence with isomorphic classes of G -bundles over Σ , so that

$$\varphi^* BG \simeq T\Sigma \oplus \mathbb{R}^0 \simeq T\Sigma,$$

meaning that, when ρ is the inclusion, Σ is endowed with a certain G -structure.

Example 4.1.2 (*Oriented and framed QFT*).

Motivation

At this moment, we only have justified the necessity of working with $\mathbf{Cob}_n^{(M,E),\mathcal{F}}$ in the abstract definition of QFT. We also need to justify why they are monoidal functor taking values in $\mathbf{Vec}_{\mathbb{C}}$. Indeed, as will be discussed now, this relies essentially in the requirement that the given definition be a generalization of Quantum Mechanics, which corresponds to the canonical description of “quantum movement” of a nonrelativistic particle. A very nice discussion about the intuitive meaning of the QFT definition also can be found in [11] as well in [8].

We initiate recalling the structure of quantum mechanics. Classical textbooks about the subject, some of them mathematically consistent and others not, are [??]. Very clear expositions are given in [12] as well in .

A *quantum mechanical system* is composed by a (complex and separable) Hilbert space $\mathcal{H}$ together a distinguished self-adjoint linear operator $H : D(H) \rightarrow \mathcal{H}$, (densely) defined

on some subspace $D(H) \subset \mathcal{H}$. We will refer to the parenthesis conditions as being *technicalities* or *technical conditions*.

The elements of $\mathcal{H}$ are called *quantum states* of the system, so that the whole $\mathcal{H}$ is the *space of quantum states*. The densely defined self-adjoint operator on $\mathcal{H}$ are the *quantum observable* of the system. The distinguished observable H is the *Hamiltonian* of the system. Given a quantum state ψ_0 , their *time evolution* from an instant t_0 is the solution of the initial valued problem

$$i \frac{d\psi}{dt} = H(\psi), \quad \text{with} \quad \psi(t_0) = \psi_0,$$

where the equation corresponds to the so called (time independent/linear) Schrödinger equation. Such problem really has an unique solution, which is given by the path

$$\psi_0 : \mathbb{R} \rightarrow \mathcal{H}, \quad \text{defined by} \quad \psi_0(t) = \exp(i(t - t_0)H). \quad (4.1)$$

How the exponential of an operator is defined always in the whole space (even if the exponentiated operator does not), we have a globally defined flow $\phi^0 : \mathbb{R} \times \mathcal{H} \rightarrow \mathcal{H}$, such that $\phi^0(t, \psi_0) = \psi_0(t)$. The flows defines, indeed, a homomorphism between the additive group of real numbers to the endomorphism group of $\mathcal{H}$, which is translated in terms of a one-parameter family $\phi_t^0 : \mathcal{H} \rightarrow \mathcal{H}$ of globally defined unitary operators.

Now, varying the initial instant t_0 we get a map $U : \mathbb{R} \times \mathbb{R} \rightarrow \mathcal{H}$, which to any (t', t) assign the quantum state $\phi_{t'}^t(\psi_t)$, which can be described in terms of a two-parameter family of unitary operators such that

$$U(t'', t') \circ U(t', t) = U(t'', t) \quad \text{and} \quad U(t, t) = id_{\mathcal{H}}, \quad (4.2)$$

where the first condition follows directly from the group property of any $\phi_t^{t'}$ and by the fact that the Schrödinger equations is autonomous (because the Hamiltonian is supposed time independent). The second, of course, follows from $\phi_t^t = id_{\mathcal{H}}$ for any t .

On the other hand, by the Stone's theorem, any smooth one-parameter family ϕ_t of globally defined unitary operators are of the form (4.1) for some self-adjoint operator H . So, we can characterize the time evolution of a quantum mechanical system as being given by a rule U which assign to any instant t a Hilbert space $U(t)$ (which, in the above description are all the same) and to any interval $[t, t']$ a globally defined unitary operator

$$U([t, t']) : U(t) \rightarrow U(t')$$

satisfying the conditions (4.2), which can put in the form

$$U([t', t''] \circ [t, t']) = U([t', t'']) \circ U([t, t']) \quad \text{and} \quad U([t, t]) = id_{\mathcal{H}}$$

if we simply write

$$[t', t''] \circ [t, t'] = [t'', t].$$

Now, observe that any interval is a cobordism between their extremities, so that U is nothing more than a functor $U : \mathbf{ConCob}_0^{\text{Riem}} \rightarrow \mathbf{Hilb}$, defined on the geometric category of oriented cobordisms between connected 0-manifolds and taking values in the category of complex Hilbert spaces and unitary globally define unitary operators! We have a natural forgetful functor

$$\mathbf{Hilb} \rightarrow \mathbf{Vec}_{\mathbb{C}}$$

which forgot what we had called *technicalities*. So, we can see a quantum mechanical system as being simply a functor from $\mathbf{ConCob}_0^{\text{Riem}}$ to $\mathbf{Vec}_{\mathbb{C}}$.

Now, the question is: *why restrict only to the connected case?* A non connected 0-dimensional manifold is simply a discrete set with more than an unique element. Any connected component (in this case, any element) corresponds to a 0-brane (that is, a particle). So, a cobordism between two of such manifolds will be simply a disjoint union of circles and intervals, meaning that the time evolution of some particle is totally independent of the others.

Happens that, in quantum mechanics, is postulated that, if $\mathcal{H}_1$ and $\mathcal{H}_2$ are the quantum space of states of two independent particles, then the space of states of the system describing both particles simultaneously is identified with $\mathcal{H}_1 \otimes \mathcal{H}_2$. Consequently, we can extend the functor U to the whole category $\mathbf{Cob}_1^{\text{Riem}}$ putting

$$U(t \sqcup t') \simeq U(t) \otimes U(t') \quad \text{and} \quad U(\emptyset) = \mathbb{C},$$

which means simply that U is a monoidal functor. Therefore, *the given definition of QFT really is a natural generalization of the time evolution of quantum mechanical systems without technicalities for arbitrary quantity of particles.*

Unitary Theories

In the last subsection we see that the given definition of quantum field theory is a direct generalization of the time evolution of quantum mechanical system *forgotten the technicalities*. So, we ask: what is the role of such technicalities in the structure of the theory to be described?

In order to answer such question, we first observe that the typical quantum mechanical systems (in which technicalities are considered) are described on *infinite dimensional* Hilbert spaces. On the other hand, as we will show below, if we forgot the technicalities, then the space of quantum states must be necessarily *finite dimensional*! But the technicalities are composed essentially by unitary time evolution, so that *the requirement that the time evolution be unitary is strict related with the fact that the standard quantum mechanical systems holds in infinite dimensional spaces.*

We observe that, if we are working with oriented cobordisms, then for any given $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ we can bought another $\Sigma^\dagger : \Sigma_1 \rightarrow \Sigma_0$ simply inverting the orientation. In particular, we can talk about *unitary cobordisms* as being those such that $\Sigma \circ \Sigma^\dagger = id_{\Sigma_1}$ and $\Sigma^\dagger \circ \Sigma = id_{\Sigma_0}$. Therefore, we can also talk about *unitary QFTs*, meaning that we have $U(\dots\dots\dots)$

- why not work on unitary theories? We give two justifications:
 - if there is no topology change, then the evolution is unitary. So, if the time evolution is not necessarily unitary, then we can have topology change. Wormholes solutions (instead being generally being formed by exotic matter) are important in the modern study of gravity and they need to topology change;
 - Thanks to Folklore theorem, for $n \leq 2$ the reciprocal is also valid. This means, for example, that for topological quantum theories of strings, there is topology change iff the time evolution is not necessarily unitary. But the “topology change” describes crucially the perturbative aspects of the theory.
 - DISCUTIR SOBRE TEORIAS COM EXISTÊNCIA DE BURACOS DE MINHOCA E A PERDA DE PREVISIBILIDADE (PELO TEOREMA DE GEROCH, ESPAÇO GLOBALMENTE HIPERBÓLICO É CILINDRO E, PORTANDO, NÃO TEM MUDANÇA TOPOLOGÓGICA).
- this motivate the maximum that *unitary evolution should not be an additional requirement, but a consequence of restrictions on the structure of the theory.*
- To end this subsection, let show that, if $U : \mathbf{Cob}_n^{SO(n)} \rightarrow \mathbf{Vec}_{\mathbb{C}}$ is an oriented QFT, then the vector space $U(\Sigma_0)$ assigned to any n -manifold Σ_0 is finite dimensional. Indeed, we can see Σ_0 as cobordisms $\emptyset \rightarrow \Sigma_0$ and $\Sigma_0 \rightarrow \emptyset$. Particularly, $\Sigma_0 \times I$ can be seen as pairings $ev_0 : \Sigma_0 \sqcup \Sigma_0 \rightarrow \emptyset$ and $coev_0 : \emptyset \rightarrow \Sigma_0 \sqcup \Sigma_0$. Applying U , we get maps

$$\Sigma_0 \rightarrow \Sigma_0 \sqcup \Sigma_0 \sqcup \Sigma_0 \rightarrow \Sigma_0 = id_{\Sigma_0}$$

$$U(ev_0) : V \otimes V \rightarrow \mathbb{C} \quad \text{and} \quad U(coev_0) : \mathbb{C} \rightarrow V \otimes V$$

tensoring with V we get a map

$$V \rightarrow V \otimes V \otimes V \rightarrow V,$$

which is the identity map.

- proof of the finite dimensi show simply that $\mathbb{S}^1$ is *dualizable*.

4.2 Structure

- theories with boundary
- theories with defect

Geometric Theories

Recall that in certain classical theories of p -branes, the worldvolume Σ have certain geometric structures which not corresponds to merely internal degrees of freedom. For example, this is the case of conformal theories and theories of gravity in some ambient with boundary. Particularly, when Σ is a cobordism between Σ_0 and Σ_1 , we generally suppose that not only Σ have geometric structures $\mathcal{F}$, but also the initial and final states Σ_0 and Σ_1 , say given by $\mathcal{F}_0$ and $\mathcal{F}_1$, so that $\mathcal{F}|_{\Sigma_0} \simeq \mathcal{F}_0$ and $\mathcal{F}|_{\Sigma_1} \simeq \mathcal{F}_1$. This can be understanding saying that the geometric structure of the initial Σ_0 is preserved along the time evolution.

The previous definition of QFT does not take into account such geometric cases. It is exactly for this reason that we call them of *topological quantum field theories*. We would like to extends such definition so that to take into account these geometric situations. The idea is build categories of *geometric cobordisms* $\mathbf{Cob}_n^{(M,E,\mathcal{F})}$ and define a *quantum field theory of type* $(M, E, \mathcal{F})$ as being a symmetric monoidal functor

$$U : \mathbf{Cob}_n^{(M,E,\mathcal{F})} \rightarrow \mathbf{Vec}_{\mathbb{C}}.$$

For example, when $\mathcal{F} = \text{SRiem}$ corresponds to semi-riemannian metrics, we can define the category $\mathbf{Cob}_n^{(M,E,\text{SRiem})}$ as being the subcategory of $\mathbf{Cob}_n^{(M,E)}$, whose n -manifold objects comes endowed with a semi-riemannian metric, as well the morphisms are given by *semi-riemannian cobordisms* $(\Sigma, g) : (\Sigma_0, g_0) \rightarrow (\Sigma_1, g_1)$, such that $g|_{\Sigma_i} = g_i$. The correspondent quantum theories are called *semi-riemannian quantum field theories*. Similarly we can define the the category $\mathbf{Cob}_n^{(M,E,\text{Conf})}$ when $\mathcal{F} = \text{Conf}$ is the conformal structure, so that we can talk about *conformal quantum field theories* in the same original spirit of Segal [10].

The abstract and general definition of $\mathbf{Cob}_n^{(M,E,\mathcal{F})}$ (and, therefore, of more abstract quantum field theory of certain geometric type) is given in [9]. There, the abstract notion of “geometric structure $\mathcal{F}$ ” is defined in terms of sheaf-theoretic perspective, we only refer the reader to the above reference for details.

- Observation: we can also look to geometric QFT’s taking values in other monoidal category rather than $\mathbf{Vec}_{\mathbb{C}}$. The main examples of CFT’s takes values not on $\mathbf{Vec}_{\mathbb{C}}$, but in their “stable version”: the monoidal category of chain complex.
- OBSERVAÇÃO: EM GERAL SE PRECISA FAZER RESTRIÇÕES PARA QUE “COBORDISMOS GEOMÉTRICOS” FAÇAM SENTIDO. POR EXEMPLO, NO CASO LORENTZIANO UMA CONDIÇÃO SUFICIENTE É PEDIR HIPERBOLICIDADE GLOBAL.
- comment about SFT

Supersymmetric Theories

Conformal Theories

- vd

4.3 Classification

Folklore

The first nontrivial examples happens in the context of 2-dimensional oriented QFTs. Indeed, in such context there is a theorem, generally referred as the *Folklore theorem*, which gives a complete classification of such QFTs. More precisely, it states that to giving a 2-dimensional oriented QFT is the same that giving a finite dimensional commutative Frobenius algebra. Therefore, giving examples of Frobenius algebras we get examples of QFTs. We will explain briefly the idea of the proof. Details can be founded in [??].

We say that a category $\mathbf{C}$ is *generated* by a set B of morphisms (or that B is a *monoidal basis* for $\mathbf{C}$) when any other morphism can be written as a composition of them. Happens that, if the category $\mathbf{C}$ is generated by B , then any functor $F : \mathbf{C} \rightarrow \mathbf{C}'$ is completely determined by their action of the elements of B (and also in their correspondents domain and codomain). Now, let $f : X \rightarrow Y$ an arbitrary morphism. Then, being B a basis, we can write $f = \circ_i f_i$ for certain $f_i \in B$, so that

$$F(f) = F(\circ_i f_i) \simeq \circ_i F(f_i),$$

showing that F really is totally determined by their action on B . On the other hand, as will be explained in the end, using the classification theorem for compact oriented 2-manifolds and Morse theory, we can see that $(\mathbf{Cob}_2^{SO(n)}, \sqcup)$ is generated by the following cobordisms:

$$\eta : \emptyset \rightarrow \mathbb{S}^1, \quad \mu : \mathbb{S}^1 \sqcup \mathbb{S}^1 \rightarrow \mathbb{S}^1, \quad \delta : \mathbb{S}^1 \rightarrow \mathbb{S}^1 \sqcup \mathbb{S}^1 \quad \text{and} \quad \epsilon : \mathbb{S}^1 \rightarrow \emptyset. \quad (4.3)$$

Therefore, any $U : \mathbf{Cob}_2^{SO(n)} \rightarrow \mathbf{Vec}_{\mathbb{C}}$ is totally determined by their action on such cobordisms, which gives a (finite dimensional, as discussed later) vector space $V = U(\mathbb{S}^1)$, together linear maps

$$\eta : \mathbb{C} \rightarrow V, \quad \mu : V \otimes V \rightarrow V, \quad \delta : V \rightarrow V \otimes V \quad \text{and} \quad \epsilon : V \rightarrow \mathbb{C}.$$

Happens that the below diagrams involving η and μ are commutative, so that U gives analogous commutative diagrams in $\mathbf{Vec}_{\mathbb{C}}$ for V . Recalling that linear maps $V \otimes V \rightarrow V$ are just the same that bilinear maps $V \times V \rightarrow V$ we see that the first of such diagrams say simply that V is a complex associative algebra with respect to the product μ . Furthermore, linear maps $\mathbb{C} \rightarrow V$ are totally determined by their images on $1 \in \mathbb{C}$. With this

identification, the second diagram say that the algebra V is unital with unity $\eta(1)$.

$$\begin{array}{ccc}
 \mathbb{S}^1 \sqcup \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xrightarrow{\mu \sqcup id} & \mathbb{S}^1 \sqcup \mathbb{S}^1 \\
 \downarrow id \sqcup \mu & & \downarrow \mu \\
 \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xrightarrow{\mu} & \mathbb{S}^1
 \end{array}
 \qquad
 \begin{array}{ccccc}
 \emptyset \sqcup \mathbb{S}^1 & \xrightarrow{\eta \sqcup id} & \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xleftarrow{id \sqcup \eta} & \mathbb{S}^1 \sqcup \emptyset \\
 \searrow \simeq & & \downarrow \mu & & \swarrow \simeq \\
 & & \mathbb{S}^1 & &
 \end{array}$$

On the other hand, the maps δ and ϵ satisfy totally dual commutative conditions, as presented in the below diagrams. So, U maps into analogous diagrams in $\mathbf{Vec}_{\mathbb{C}}$, which means that V also has a dual algebraic structure with respect to δ and ϵ , usually called *counital coassociative coalgebra*.

$$\begin{array}{ccc}
 \mathbb{S}^1 \sqcup \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xleftarrow{\delta \sqcup id} & \mathbb{S}^1 \sqcup \mathbb{S}^1 \\
 \uparrow id \sqcup \delta & & \uparrow \delta \\
 \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xleftarrow{\delta} & \mathbb{S}^1
 \end{array}
 \qquad
 \begin{array}{ccccc}
 \emptyset \sqcup \mathbb{S}^1 & \xleftarrow{\epsilon \sqcup id} & \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xrightarrow{id \sqcup \epsilon} & \mathbb{S}^1 \sqcup \emptyset \\
 \searrow \simeq & & \uparrow \delta & & \swarrow \simeq \\
 & & \mathbb{S}^1 & &
 \end{array}$$

The maps (4.3) satisfy further conditions. Indeed, they also satisfy conditions which will make (V, μ, η) a commutative algebra and, dually, (V, δ, ϵ) a cocommutative algebra. Finally, they satisfy certain compatibility conditions, which can be paraphrased in terms of the commutativity of the below diagrams. Certainly, analogous condition will be satisfied by V . But, when viewed together, all such conditions say simply that V is a *commutative Frobenius algebra*. Therefore, any 2-dimensional oriented QFT is totally determined by the commutative Frobenius algebra associated to $\mathbb{S}^1$.

$$\begin{array}{ccccc}
 & & \mathbb{S}^1 \sqcup \mathbb{S}^1 & & \\
 & \searrow \delta \sqcup id & \downarrow \mu & \swarrow id \sqcup \delta & \\
 \mathbb{S}^1 \sqcup \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xrightarrow{id \sqcup \mu} & \mathbb{S}^1 \sqcup \mathbb{S}^1 & \xleftarrow{\mu \sqcup id} & \mathbb{S}^1 \sqcup \mathbb{S}^1 \sqcup \mathbb{S}^1
 \end{array}$$

Now, let is prove that the cobordisms (4.3) really generate the whole category $\mathbf{Cob}_2^{SO(n)}$. Indeed, let $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ an arbitrary cobordism. Both Σ_0 and Σ_1 are 1-dimensional compact manifold without boundary, so that by the classification of 1-manifolds we can write $\Sigma_0 \simeq \sqcup_{i \in I} \mathbb{S}^1$ and $\Sigma_1 \simeq \sqcup_{j \in J} \mathbb{S}^1$ for certain I and J . We have two cases: or both I and J are empty, or some of them do not is. Such cases will be studied separately.

1. The first corresponds to $\Sigma : \emptyset \rightarrow \emptyset$, so that Σ is a compact oriented 2-manifold without boundary. Therefore, by the classification of 2-manifolds, it is diffeomorphic to a sphere or to a connected sum of torus. Now, pick out the height function as a Morse function on Σ . It has exactly $2g + 2$ critical points, where g is the genus. The lower critical point is a source, as well as the upper is a sink. Therefore, they indexes are equal to zero and two, respectively. The others are all saddles and, therefore, have index equal to one. Therefore, Σ is diffeomorphic to the composition

$$\emptyset \longrightarrow \mathbb{S}^1 \longrightarrow \mathbb{S}^1 \sqcup \mathbb{S}^1 \longrightarrow \mathbb{S}^1 \longrightarrow \dots \longrightarrow \mathbb{S}^1 \longrightarrow \mathbb{S}^1 \sqcup \mathbb{S}^1 \longrightarrow \mathbb{S}^1 \longrightarrow \emptyset$$

being, therefore, generated by the cobordisms (4.3).

2. In the second case, $\Sigma : \sqcup_i \mathbb{S}^1 \rightarrow \sqcup_j \mathbb{S}^1$ is compact, oriented, but has non empty boundary. Therefore, we cannot use directly the classification of 2-manifolds. But observe that if we consider the composition

$$\emptyset \longrightarrow \sqcup_i \mathbb{S}^1 \xrightarrow{\Sigma} \sqcup_j \mathbb{S}^1 \longrightarrow \emptyset ,$$

then it is a 2-manifold without boundary, so that we can apply the classification theorem as in the previous case and then show that Σ also is composed by (4.3).

Examples

Example 4.3.1 (*ordinary cohomology*).

Example 4.3.2 (*quantum cohomology*).

Example 4.3.3 (*index on K-theory*).

TV Construction

Examples

- examples of modular categories.

FRS-Theorem

Chapter 5

Quantization

At this moment, the only concrete nontrivial examples of QFTs that we know are those provided from Frobenius algebras by folklore. On the other hand, in the last chapter we had build many examples of classical theories. Therefore, in order to get new examples of quantum theories, we can try to get “quantum analogous” of some classical theory. Any map associating quantum theories to a given classical theory (as present in the below) is called a *quantization procedure*.

Having developed such procedures, we will be able to talk about *sigma model quantum field theories*. These will be just the QFT obtained from quantization of some sigma model.

$$\text{classical theories} \xRightarrow{\text{quantization}} \text{quantum theories}$$

Presently, there are many proposes but there is no consensus about the correct definition of quantization [??]. For instance, one of the Millennium Problems of the Clay Institute of Mathematics concerns the construction of a quantum version of Yang-Mills theory for any compact simple Lie group G satisfying desirable properties [??].

We started this chapter giving a precise definition of what quantization can be and presenting a strategy to build quantization satisfying the given definition. Such strategy will makes use of the concept of *path integral*, which, unfortunately, presently is also obscure. We discuss how to get some felling about path integrals.....

5.1 Definition

In order to give the definition of quantization, let us recall that a classical theory of p -branes is obtained

1. fixing a cobordism $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ between p -manifolds, which may have additional geometric structure and internal degrees of freedom;
2. giving a *phase space* $\text{Fields}(\Sigma)$, which is a set of field configurations. Such set generally has additional structure. For instance, it can be.....

3. giving an *action functional* $S : \text{Fields}(\Sigma) \rightarrow \mathbb{R}$, possibly being a map with additional regularity. For example, ;
4. looking to a distinguished *covariant phase space* $\text{Cov}(\Sigma) \hookrightarrow \text{Fields}(\Sigma)$. For example, when S stay in a category in which we can take derivatives, it is generally composed by the states which satisfy the Euler-Lagrange equation $dS = 0$.

On the other hand, by their probabilist nature, quantum theories are defined not only in a *specific* worldvolume Σ , but on any of them. So, the main idea not is consider a quantization procedure as being an effective rule assign to any specific classical theory a correspondent quantum version, but as a rule which assign to any “class” of classical theories of type (M, E) a correspondent quantum theory of the same type.

More precisely, a *class of classical theories* is composed by three functions Fields , as well as S_- and Cov . The first of them assign to any cobordism Σ a set $\text{Fields}(\Sigma)$, possibly with more structure, corresponding to the phase space associated with the abstract motion Σ . The second, in turn, assign to any Σ an action functional $S_\Sigma : \text{Fields}(\Sigma) \rightarrow \mathbb{R}$. Finally, the third select, for any cobordism, a distinguished subspace $\text{Cov}(\Sigma)$ of the phase space, meaning the covariant phase space.

The main examples to have in mind are those given by sigma-model theories internal to some suitable context $\mathbf{H}$, which in general is a cohesive ∞ -topos (for instance, the most important classical theories arises in this way for $\mathbf{H} = \text{Smooth}\infty\text{Stack}$ or some other nice category obtained from it). In such cases, the phase space map extends to a monoidal functor $\text{Fields} : \mathbf{Cob}_p \rightarrow \text{Span}(\mathbf{H})$, as well as the action $S : \text{Fields} \rightarrow \mathbb{R}$ is simply a lifting of such monoidal functor to $\mathbf{H}/\mathbf{B}\mathbb{R}$. The covariant phase space, in turn, is obtained forcing the minimization of S . On the other hand, instead of such kind of classical theories are the most prominents, in principle can be exists others that are not internal sigma-models.

We have a natural category of quantum theories of a given type. Indeed, being quantum theories given by monoidal functors, the required category is just the full functor subcategory

$$\text{Qnt}_p^{(M,E)} \hookrightarrow \text{Func}(\mathbf{Cob}_p^{(M,E)}; \mathbf{C}),$$

whose morphisms are just natural transformations. Similarly, we can define a category $\text{Clas}_p^{(M,E)}$ of classes of classical theories of the prescribed type. Indeed, a morphism between two of such classes, is simply a function f_- which preserve all the involved structures, in the sense that:

1. *preservation of phase spaces*: for any Σ , we have a map $f_\Sigma : \text{Fields}(\Sigma) \rightarrow \text{Fields}'(\Sigma)$, possibly with more regularity (say smooth, if the phase spaces has smooth structures);
2. *preservation of actions*: for any Σ , we have $S_\Sigma = S'_\Sigma \circ f_\Sigma$;
3. *preservation of covariant phase space*: if φ is a distinguished state, then $f_\Sigma(\varphi)$ satisfy is a distinguished state.

All of such conditions can be condensed in the requirement that the first below diagram commutes for any abstract motion. For example, when both theories are sigma-models internal to some context $\mathbf{H}$,

$$\begin{array}{ccc}
 \text{Fields}^0(\Sigma) & \xrightarrow{f_\Sigma} & \text{Fields}'^0(\Sigma) \\
 \downarrow & & \downarrow \\
 \text{Fields}(\Sigma) & \xrightarrow{f_\Sigma} & \text{Fields}'(\Sigma) \\
 & \searrow S_\Sigma \quad \swarrow S'_\Sigma & \\
 & \mathbb{R} &
 \end{array}$$

Now, it seems natural to define a *quantization* as being simply a functor between the classical category to the quantum category:

$$\mathcal{Q} : \text{Clas}_p^{(M,E)} \rightarrow \text{Qnt}_p^{(M,E)}.$$

Existence and Uniqueness

What can be said about the existence of quantization in the the sense of the previous definition? In the next subsection, we will present a strategy to build a type of quantization called *quantization by path integrals*. As suggested by the name, such construction involves an object called path integral, whose precise definition presently it is not well understand. So, in this general context, what will be done there is to give certain insight about what means to quantize a classical system.

On the other hand, in certain particular cases the path integrals can be effectively build and the quantization by path integrals has a rigorous meaning. More precisely, there is a subcategory $\text{PathInt} \subset \text{Clas}_p^{(M,E)}$, formed simply by the classical theories for that the path integral makes sense, in which we have a quantization

$$\mathcal{Q} : \text{PathInt} \rightarrow \text{Qnt}_p^{(M,E)}.$$

There exists many other projects that try build quantization avoiding the notion of path integral. For example, there is the *geometric quantization*, which build a quantization of classical theories in which $\mathcal{C}_\Sigma^0$ comes endowed with a natural symplectic structure, as well as *deformation quantization* (which actually is not a quantization in the sense of the given definition) [??],..... and the modern *quantization by the A-model* [??]. Similar than the path integral quantization, any of such attempts are defined only on a subcategory. Therefore, like in functional analysis (where we generally work with self-adjoints operators defined on a subspace), it seems to be natural to associate a domain $D(\mathcal{Q}) \subseteq \text{Clas}_p^{(M,E)}$ to any attempt to quantization. Then, we can ask about the existence of extensions, as

present below.

$$\begin{array}{ccc}
 & & \text{Qnt}_p^{(M,E)} \\
 & \nearrow \mathcal{Q} & \uparrow \text{---} \\
 D(\mathcal{Q}) & \xrightarrow{i} & \text{Clas}_p^{(M,E)}
 \end{array}$$

Generally extension problems admits obstructions. For instance, in topology such obstructions are given by *obstruction theory* and, for example, prevent that any fiber bundle have a global section. On the other hand, in function analysis we also have obstructions to extend a self-adjoint operator to the whole space. Indeed, by the Hellinger-Toeplitz theorem, such extension exists only if the operator is bounded [??]. Therefore, it is natural to believe that obstructions also be present in the extension of quantization. For example, such obstructions really exists in the context of geometric quantization, being given by the Van Hove-Groenewold-Gotay examples (see [2, 3, 4, 5, 6] for the referred examples as well as [7] and [??] for good discussions). Here, the obstructions are given on morphisms, in the sense that, for a fixed symplectic structure ω on $\text{Cov}(\Sigma)$, not all automorphisms $\text{Cov}(\Sigma) \rightarrow \text{Cov}(\Sigma)$ can be quantized.

We observe that the general extension problem to a given quantization is presently an open problem. For example, for the path integral quantization it corresponds to the problem of giving a precise definition of path integral in a general context. But besides the extension problem we can also analysis the *uniqueness problem* of quantization procedures. More precisely, given two quantizations, say $\mathcal{Q}$ and $\mathcal{Q}'$ defined on their respective domain $D(\mathcal{Q})$ and $D(\mathcal{Q}')$, we can ask if in some sense they are equivalent when restricted to some subcategory contained in the intersection $D(\mathcal{Q}) \cap D(\mathcal{Q}')$. In this general context, currently it also is an open problem.

On the other hand, restricting to the context of classical mechanics (i.e looking to the subcategory of all class of classical theories such that $p = 1$, the ambient space M is Riemannian, the phase spaces $\text{Fields}(\Sigma)$ are simply the set of paths $\varphi : \Sigma \rightarrow M$ and the actions S_Σ are given by a Lagrangian that depends of φ and $D\varphi$), then geometric quantization, path integral quantization and deformation quantization can be applied producing parts of a same theory: quantum mechanics.

Inverse Problem

- inverse problem to quantization. Fix a quantization approach. Then, given a quantum theory, obtain some classical theory whose quantizable version is the given quantum theory.

Example 5.1.1 (*sigma model QFT from invariants*).

- twisted A-model as being “essentially” the inverse problem for the quantum cohomology. The problem is that the theory don’t is topological (and, actually, also don’t is conformal). Idea: pass from higher geometry and see it as a higher (or extended) topological quantum field theory.

5.2 Path Integral

Here the quantization by path integrals will be discussed. We initiate observing that, in order to build a quantization procedure we need to associate a monoidal functor $\mathcal{Q}_S : \text{Qnt}_p^{(M,E)} \rightarrow \mathbf{Vec}_{\mathbb{C}}$ to any class of classical system with action S_- and a natural transformation to any morphism of classical systems. We start focusing on the functors $\mathcal{Q}_S$. To define them, we need to assign a vector space $\mathcal{Q}_S(\Sigma_0)$ to any p -manifold Σ_0 , as well as a linear operator

$$\mathcal{Q}_S(\Sigma) : \mathcal{Q}_S(\Sigma_0) \rightarrow \mathcal{Q}_S(\Sigma_1)$$

to any cobordism $\Sigma : \Sigma_0 \rightarrow \Sigma_1$, such that compositions and identities are preserved and disjoint are mapped on tensor products:

$$\mathcal{Q}_S(\Sigma_0 \sqcup \Sigma'_0) \simeq \mathcal{Q}_S(\Sigma_0) \otimes \mathcal{Q}_S(\Sigma'_0). \quad (5.1)$$

So, first of all we are looking to a way to assign complex vector spaces to p -manifolds, which will represent the space of quantum states of a p -brane. In the context of quantum mechanics, these quantum spaces are generally given by certain L^2 -space, so that the idea is to consider $\mathcal{Q}_S(\Sigma_0)$ as being some L^2 -space. Now we observe that, if Σ_0 is *oriented*, then there are natural L^2 -spaces that can be associated to it. Indeed, fixed a hermitean bundle $E_0 \rightarrow \Sigma_0$, its space of sections $\Gamma(E_0)$ carries a natural bilinear map

$$\langle -, - \rangle : \Gamma(E_0) \times \Gamma(E_0) \rightarrow C^\infty(\Sigma_0; \mathbb{C}), \quad \text{given by} \quad \langle \psi, \psi' \rangle(x) = \langle \psi(x), \psi'(x) \rangle_x.$$

Therefore, being Σ_0 oriented, we can use its volume form ω together the above map to define the space $L^2(\Sigma_0)$ of square-integrable sections of E_0 with respect to the inner product

$$(\psi, \psi') = \int_{\Sigma_0} \langle \psi, \psi' \rangle \cdot \omega,$$

motivating us to put $\mathcal{Q}_S(\Sigma_0) = L^2(\Sigma_0)$. Now, in order to see that the condition (5.1) really is satisfied, we need associate a hermitean bundle over $\Sigma_0 \sqcup \Sigma'_0$. To do this, recall that almost complex bundles are obtained from real bundles reducing the structural group $GL(2n; \mathbb{R})$ to $GL(n; \mathbb{C})$, as well as hermitean bundles are integrable reductions to $U(n)$. So, let

$$\varphi_{ij} : U_i \cap U_j \rightarrow U(n) \quad \text{and} \quad \varphi'_{ij} : U'_i \cap U'_j \rightarrow U(n')$$

be respective coordinates of $E_0 \rightarrow \Sigma_0$ and $E'_0 \rightarrow \Sigma'_0$. Then define new maps ϕ_{ij} from certain intersection of open sets on $\Sigma_0 \sqcup \Sigma'_0$ to $U(n) \oplus U(n')$ putting

$$\phi_{ij}(x) = \varphi_{ij}(x) \oplus I_{n'} \quad \text{or} \quad \phi_{ij}(x) = I_n \oplus \varphi'_{ij}(x)$$

according to $x \in \Sigma_0$ or $x \in \Sigma'_0$. Such maps evidently satisfy the Čech cocycle condition, because it is satisfied by φ_{ij} and by φ'_{ij} . Therefore, they define a bundle over $\Sigma_0 \sqcup \Sigma'_0$ structured by $U(n) \oplus U(n')$, which is almost hermitean, because $U(n) \oplus U(n') \subset U(n+n')$, and indeed integrable (because E_0 and E'_0 are integrable). Such bundle is simply $E_0 \oplus E'_0$, so that we take

$$\mathcal{Q}_S(\Sigma_0 \sqcup \Sigma'_0) := L^2(\Gamma(E_0 \oplus E'_0)).$$

Now, being the section functor Γ and the L^2 -map monoidal, we see that the condition (5.1) really is satisfied:

$$L^2(\Gamma(E_0 \oplus E'_0)) \simeq L^2(\Gamma(E_0) \oplus \Gamma(E'_0)) \simeq L^2(\Gamma(E_0)) \otimes L^2(\Gamma(E'_0))$$

To define the functor $\mathcal{Q}_S$ on morphisms, for any given cobordism $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ we need to build an operator

$$\mathcal{Q}_S(\Sigma) : L^2(\Sigma_0) \rightarrow L^2(\Sigma_1)$$

between L^2 -spaces. Natural operators between spaces of integrable functions are those given by *integral transforms*. So, the main idea is to put something like this

$$\mathcal{Q}_S(\Sigma)(\psi)(x) = \int K(x, y) \psi(y) d\mu,$$

where the kernel K should be a function with domain on $\Sigma_1 \times \Sigma_0$ defined in terms of S . But for any $y \in \Sigma_0$ the evaluation $\psi(y)$ takes values in a different space (the fiber of $\mathbb{L}_0$ on y). Therefore, for any x the map $K(x, -)$ should be a complex valued function and the above integration should be a certain kind of *fiber integration*.

To define K recall that quantum theories are probabilistic. This means that, differently of classical theories (in that the only possible configurations $\varphi : \Sigma \rightarrow M$ are which minimize the action S_Σ), we should take into account any of such configurations, but with a “probabilistic weight” associated to them, which must be determined by the classical action S_Σ . Particularly, we expect that such probabilities oscillate in the neighborhood of the classically observed configurations, so that a nice choice for the weight of a given $\varphi : \Sigma \rightarrow M$ is $e^{iS_\Sigma[\varphi]}$. Furthermore, $K(x, y)$ should be a certain kind of “sum over all weights”. But the weights are labelled by the space of configurations that generally is not discrete. So we are summing over a continuum set, motivating us to write

$$K(x, y) = \int e^{iS_\Sigma[\varphi]} D\varphi. \quad (5.2)$$

Finally, to conclude the construction of the path integral quantization, we need to assign a natural transformation $\mathcal{Q}_f : \mathcal{Q}_S \Rightarrow \mathcal{Q}_{S'}$ to any morphism of classical theories. Therefore, for each Σ_0 we need build a map $\mathcal{Q}_{f_{\Sigma_0}}$ such that the following diagram commutes

for any Σ

$$\begin{array}{ccc}
 \mathcal{Q}_S(\Sigma_0) & \xrightarrow{\int e^{iS_\Sigma[\varphi]} D_\varphi} & \mathcal{Q}_S(\Sigma_1) \\
 \mathcal{Q}_{f_{\Sigma_0}} \downarrow & & \downarrow \mathcal{Q}_{f_{\Sigma_1}} \\
 \mathcal{Q}_{S'}(\Sigma_0) & \xrightarrow{\int e^{iS'_\Sigma[\varphi']} D'_{\varphi'}} & \mathcal{Q}_{S'}(\Sigma_1)
 \end{array}$$

To do this, assume for some moment that the notion of path integral really is well defined and that it behaves as an usual integration (we will discuss more about this in the sequence). Now, observe that the definition of $\mathcal{Q}_S(\Sigma_0)$ depends only of Σ_0 and of the choice of a hermitean bundle on Σ_0 . So, such space does not depends explicitly of the theory S and, therefore, we can take $\mathcal{Q}_S(\Sigma_0) = \mathcal{Q}_{S'}(\Sigma_0)$. Furthermore, being f a morphism between class of classical theories, we have the equality $S = S' \circ f$, so that

$$\int \exp(iS_\Sigma[\varphi]) D_\varphi = \int \exp(iS'_\Sigma[f(\varphi)]) D'_{f(\varphi)}.$$

But recall the pullback formula for differential forms

$$\begin{aligned}
 (f^*\omega)_x(v_1, \dots, v_n) &= \omega_{f(x)}(Df_x(v_1), \dots, Df_x(v_n)). \\
 &= S(f(x))\omega'_{f(x)}(Df_x(v_1), \dots, Df_x(v_n)),
 \end{aligned}$$

where the last equality holds when $\omega = g \cdot \omega'$. The right side seems like

$$\exp(iS'_\Sigma[f(\varphi)]) D'_{f(\varphi)}, \quad \text{with} \quad g \cdot \omega' = \exp(iS'_\Sigma) \cdot D',$$

motivating us to write something like this

$$f_\Sigma^*(\exp(iS'_\Sigma) \cdot D') = \exp(iS_\Sigma) \cdot D.$$

Therefore,

$$\int \exp(iS_\Sigma) \cdot D = \int f_\Sigma^*(\exp(iS'_\Sigma) \cdot D') = \int \exp(iS'_\Sigma) \cdot D'$$

so that we can put...

Problems and Particular Cases

The standard way to try to interpret the path is as an effective integration operation. But this cannot be done in the general case. Indeed, such integration is done over $\text{Fields}(\Sigma)$ which in generally contains spaces like $\text{Map}(\Sigma; M)$ or $\Gamma(E)$, so that D_φ should be a measure in this spaces. Happens that, in general such spaces are very bad behaved by the viewpoint of measure/integration theory, so that presently there are no canonical way to define the

measure D_φ . Reviews about the difficulties and possibly attempts to a rigorous definition are [??].

On the other hand, in some very special situations the measure D_φ can be canonically defined and the expression (5.2) can be effectively interpreted as an integration. E.g, when we are working with 0-branes moving in some Riemannian background subjected to some $U(1)$ -gauge force. Indeed, on such conditions, $\text{Fields}(\Sigma)$ is simply the space of continuous paths in M as well as the action is given by the holomy of the $U(1)$ -connection. So we can consider the so called *stochastic* (or *Wiener*) *measure* [??]. For Lorentzian signature, restricting to the purely kinetic case and making a change of coordinates $t \mapsto it$ (called *Wick rotation*), $e^{iS_\Sigma[\varphi]}$ behaves like a Gaussian exponential and then we can consider D_φ as being the *Gaussian measure* [??].

To get some felling, let us consider the very simple case of particles moving in an oriented ambient space M along a finite interval of time $[t, t']$. Therefore, the configuration space $\text{Fields}[t, t']$ is simply the space of continuous paths $\varphi : [t, t'] \rightarrow M$ so that to quantize the system we need to define the measure D_φ and, consequently, the path integral

$$K(t, t') = \int e^{iS[\varphi]} D_\varphi. \quad (5.3)$$

In order to do this, observe that the choice of a certain $t < t_0 < t'$ allow us to replace any path φ by a point of M : its evaluation $\varphi(t_0)$ at t_0 . But, for any t_0 , such evaluation map $ev_{t_0} : \mathcal{C}_{[t, t']} \rightarrow M$ is surjective, because for each $p \in M$ there is at least one path φ such that $\varphi(t_0) = p$: the constant path on p . Therefore, we can put a measure $d\varphi(t_0) = A \cdot \omega$ in the image of ev_{t_0} (given by some rescale or normalization of the volume form of M) so that the following integral is well defined:

$$\int e^{iS[\varphi]} d\varphi(t_0) = \int_M e^{iS[\varphi]} A \cdot \omega.$$

We can think about it as being a “point approximation” to (5.3). Considering a finite partition $t < t_0 < \dots < t_n < t'$ of the interval $[t, t']$ we can take the “finite approximation”

$$\int \dots \int e^{iS[\varphi]} A^n \cdot d\varphi(t_0) \dots d\varphi(t_n) = \int_{M \times \dots \times M} e^{iS[\varphi]} A^n \cdot \omega^n,$$

where $\omega^n = \pi_0^* \omega \wedge \dots \wedge \pi_n^* \omega$ with π_i being the canonical projections. The natural strategy is then try to define (5.3) as being the continuum limit $n \rightarrow \infty$ of the finite approximation. In other words, in this approach, the measure D_φ should be the limit of $e^{iS[\varphi]} A^n \omega^n$. This is the standard way that the quantum mechanical path integral is introduced on physics books, e.g [??]. For a formalization of this approach for actions $S[\varphi]$ given by mechanical Lagrangians with suitable potentials, see [1].

Another special situation in which the path integral can be effectively interpreted as an integration is in the context of conformal (bosonic) string theory. More precisely, for any given $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ *oriented* cobordism, let

$$S_\Sigma : \text{Map}(\Sigma; M) \times \text{Met}_0(\Sigma) \rightarrow \mathbb{R}$$

be some bosonic action, where the second argument corresponds to the oriented metrics (that is, reductions to $SO(n)$). We need to compute the path integral

$$\int \int e^{iS_{\Sigma}[\varphi, g]} D_{\varphi} D_g. \quad (5.4)$$

Now, suppose that the action S_{Σ} has conformal symmetry, what means that we can pass to the quotient

$$S_{\Sigma} : \text{Map}(\Sigma; M) \times \text{Met}_0(\Sigma)/\text{Conf}(\Sigma) \rightarrow \mathbb{R}.$$

Therefore, the exponential term in (5.4) is invariant by conformal transformations. We observe that this *not* means that the whole path integral is a conformal invariant. Indeed, we also need to verify that the (actually not well defined) measure D_g also is invariant. This is about the existence of *quantum anomalies*, which will be discussed in the next section. But to the following we will suppose that D_g also is invariant (meaning that there is no *Weyl anomaly*). So, the integration $\int D_g$ is done over the moduli space of conformal structures instead of the space of all metrics. This space in general also is very bad behaved, so that there is no canonical way to define the measure D_g . But, as was discussed in (??), *in two dimensions oriented conformal structures are into 1-1 correspondence with complex structures*, which has a structure of finite dimensional orbifold.

So, under such conditions we are in fact integrating over a finite dimensional space, which means that the measure D_g can be rigorously defined and the path integral $\int D_g$ can be effectively interpreted as an integration operation. The other path integral on (5.4), say $\int D_{\varphi}$, at this point unfortunately remains problematic.

We observe that if we add fermionic degrees of freedom, working in a supersymmetric conformal string theory (here the Weyl anomaly cancels when $\dim M = 10$), then the space of fermionic configurations have a structure of supermanifold, so that in it we can define a precise notion of *Berezian integration*, meaning that the fermionic path integral.....

- fermionic anomaly

Anomalies and Orientations

- CANCELLATION OF ANOMALIES CAN BE INTERPRETATED AS THE EXISTENCE (AND CHOICE) OF AN ORIENTATION IN SOME GENERALIZED COHOMOLOGY THEORY.

Strategies

Instead of insisting to interpret the path integral (5.2) as a true integration operation, we can try to look to it as another type of entity having similar properties than a genuine integration operation. For example:

1. we try to define it as a certain kind of “pull-push” transformation, as will be explained in the next section;
2. observe that (5.2) involves an exponential term. So, another approach is to expand such exponential in Taylor’s series and then formally manipulates the terms putting them in a new format such that the sum of all terms now give something with precise mathematical meaning.....
3. nevertheless when are not well defined, integrals like (5.2) can be very useful to get insights about powerful invariant. Particularly, in some cases we can express known invariant purely in terms of some path integral and then we can use it to *define* such integrals as being the correspondent invariant. Examples are.....

• PERTUBATION THEORY AND THE RENORMALIZATION PROBLEM

Renormalization

- bosonic and supersymmetric string theories are renormalizable.
- brief idea about the Costello’s approach and the Cones’s approach.

5.3 Pull-Push

Here it will be discussed an alternative way to look to the path integral, i.e we will try to give a concrete and rigorous construction of some quantization. Recall that in the path integral quantization, the path integral appears as the kernel of the integral

$$\begin{aligned}\mathcal{Q}_S(\Sigma)(\psi)(x) &= \int K(x, y)\psi(y)d\mu, \\ &= \int \left(\int e^{iS_\Sigma[\varphi]} D_\varphi \right) \psi(y) d\mu\end{aligned}\tag{5.5}$$

which defines the form that the quantum theory $\mathcal{Q}_S$ assigned to the classical theory with action S act on the morphism $\Sigma : \Sigma_0 \rightarrow \Sigma_1$. So, in more precise terms, what we are going to do here is reinterpret (5.5). The idea is the following: as was said, (5.5) is an *integration along the fibers of some bundle over Σ_1* . Such fiber integration is a certain kind of *pushforward*. On the other hand, $\exp(iS_\Sigma[\varphi]) \cdot D_\varphi$ behaves like a differential form on $\text{Fields}(\Sigma)$ which can be pulled back. Therefore, we can think about (5.5) as being obtained by a pullback followed by a pushforward, i.e as being a *pull-push transform*.

Unfortunately, if in one hand this approach is more rigorous, in the other it lose at generality: *it applies only to the classical theories of p -branes given by sigma-models internal to some context* **H**. The construction starts recalling that in such class of theories

the phase space rule and the action functional are given by functors, as present in the first below diagram.

$$\begin{array}{ccc}
 \text{Span}(\mathbf{H}) & \xleftarrow{\mathbb{R}} & \text{Span}(\mathbf{H}/\mathbf{B}\mathbb{R}) \\
 \swarrow \text{Fields} & \uparrow S & \\
 & \mathbf{Cob}_p &
 \end{array}
 \qquad
 \begin{array}{ccc}
 \text{Span}(\mathbf{H}) & \xleftarrow{\mathbb{R}} & \text{Span}(\mathbf{H}/\mathbf{B}\mathbb{R}) \xrightarrow{e^i} \text{Span}(\mathbf{H}/\mathbf{B}U(1)) \\
 \swarrow \text{Fields} & \uparrow S & \nearrow e^{iS} \\
 & \mathbf{Cob}_p &
 \end{array}$$

Now, recall that we are trying to give a precise formulation of the path integral. Happens that in the path integral the action appears only *exponentiated*. In the classical language, the action is a rule assigning a number $S[\varphi] \in \mathbb{R}$ to any field φ , so that its exponential is a rule assigning an element $e^{iS[\varphi]} \in \mathbb{R}/\mathbb{Z}$ to any field. In the internal language, the quotient $\mathbb{R}/\mathbb{Z}$ makes sense as being the entity naturally associated to the ∞ -groupoid $\mathbb{R}/\mathbb{Z}$. For instance, such quotient is a Lie groupoid so that when $\mathbf{H}$ is the ∞ -topos of smooth ∞ -stacks, then we can consider the smooth stack associated to it.

The quotient map $e^i : \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$ induces a canonical morphism $\mathbf{H}/\mathbf{B}\mathbb{R} \rightarrow \mathbf{H}/\mathbf{B}U(1)$, where here we are writing $U(1) = \mathbb{R}/\mathbb{Z}$ in analogy to $\mathbb{R}/\mathbb{Z} \simeq U(1)$. Therefore, in the internal language the exponentiated action functional is defined as being the composition present in the second above diagram. This means that it is a morphism $e^{iS} : \text{Fields} \rightarrow U(1)$ in $\mathbf{Cat}/\mathbf{B}U(1)$. The path integral then can be defined as being a monoidal functor

$$\int : \text{Span}(\mathbf{H}/\mathbf{B}U(1)) \rightarrow \mathbf{D},$$

so that we can assign a quantum theory $\mathcal{Q}_S$ to a given classical theory with action S simply taking the composition:

$$\begin{array}{ccccc}
 \mathbf{Cob}_p & \xrightarrow{S} & \text{Span}(\mathbf{H}/\mathbf{B}\mathbb{R}) & \xrightarrow{e^i} & \text{Span}(\mathbf{H}/\mathbf{B}U(1)) \xrightarrow{\int} \mathbf{D} \\
 & & & \searrow \mathcal{Q}_S = \int e^{iS} &
 \end{array}$$

But recall that in order to compute the path integral we need to build certain measure $D\varphi$. The inexistence of such measure is exactly the guilty for the absense of a global interpretation of the path integral as a real integration procedure. Furthermore, the cases in which it can really be defined are those free of quantum anomalies. For instance, the measure can be defined for bosonic strings exactly when the theory is absent of Weyl anomalies, as well as it can be defined for fermions precisely when there are no fermionic anomaly. On the other hand, when quantum anomalies can be broken, to define the path integral we need to make certain choice, generally corresponding to the choice of an orientation in certain cohomology.

In the approach that we are developing here, it is expected a similar situation. Indeed, it is expected that there exists certain obstructions to the existence of the functor $\int$, corresponding to the presence of different quantum anomalies. If $\text{Span}^{or}(\mathbf{H}/\mathbf{B}U(1))$

.....

$$\begin{array}{ccccc}
 & & & \mathbf{D} & \\
 & & \mathcal{Q}_S = \int e^{iS} D\varphi & \uparrow \int & \\
 & & e^{iS} D\varphi & \text{Span}^{or}(\mathbf{H}/\mathbf{BU}(1)) & \\
 \mathbf{Cob}_p & \xrightarrow{S} & \text{Span}(\mathbf{H}/\mathbf{B}\mathbb{R}) & \xrightarrow{e^i} & \text{Span}(\mathbf{H}/\mathbf{BU}(1))
 \end{array}$$

Therefore, our work will be to build a category of *oriented spans* (relating it to *orientations on some generalized cohomology theory*, in such way that any internal sigma-model with action S lifts to a functor $e^{iS} D\varphi$), as well as a canonical monoidal functor $\int$ (representing the *path integral operator*) assigning to any oriented span an element of a fixed monoidal category $\mathbf{D}$ and, consequently, to any action S the correspondent quantum theory $\int e^{iS} D\varphi$.

The construction of the above structures will be sketched in the next subsection. Before to do this, let us make some observations:

1. dfgdg
2. dgdf

Linearization

Here we will discuss the construction of the path integral functor

$$\int : \text{Span}^{or}(\mathbf{H}/\mathbf{BU}(1)) \rightarrow \mathbf{D},$$

whose domain must be a category of orientations on some cohomology theory. Now recall that the most abstract generalized cohomology is the *twisted generalized cohomology*. It is simply the space of sections (or cosections) of a vector bundle induced from a G -principal bundle by a choice of action of G into a vector space V , which will be the typical fiber. So, the first step is to identify natural principal bundles and then look to some induced bundle.

Recall that the path integral will act on lifts of exponentiated action functionals. These are functors $e^{iS} : \mathbf{Cob}_p \rightarrow \mathbf{H}/\mathbf{BU}(1)$. They assign to any $(p-1)$ -manifold Σ_0 a map $\text{Fields}(\Sigma_0) \rightarrow \mathbf{BU}(1)$ in $\mathbf{H}$, as well as to any cobordism a homotopy, as present below. But, as was discussed in the section (??), the maps taking values in $\mathbf{BG}$ models precisely G -principal bundles internal to $\mathbf{H}$. Therefore, the exponentiated action functional assign a $U(1)$ -principal bundle $P(\Sigma_0)$ over $\text{Fields}(\Sigma_0)$ to any manifold Σ_0 as well as a pair of $U(1)$ -bundles $P^0(\Sigma)$ and $P^1(\Sigma)$ over $\text{Fields}(\Sigma)$ to any cobordism $\Sigma : \Sigma_0 \rightarrow \Sigma_1$. Particularly, because the construction is invariant by homotopy, we have that $P^0(\Sigma) \simeq P^1(\Sigma)$.

So, we really have natural principal bundles associated with the exponentiated action functional. They are $U(1)$ -bundles. The group $U(1)$ act naturally in $\mathbb{C}$ by multiplication, so that we also have canonical vector bundles internal to $\mathbf{H}$ associated with any action functional, which are obtained considering the pullbacks that can be formed in the second above diagram. So, next step would be consider the space of sections (or cosections), which would be a twisted cohomology. More precisely, recalling that there is a functor $\Gamma : \mathbf{H}/\mathbf{BU}(1) \rightarrow \mathbf{Gpd}_\infty$ assigning to any $\pi : X \rightarrow \mathbf{BU}(1)$ the space of sections of the complex line bundle associated to the $U(1)$ -principal bundle modeled by such map. For instance, by the discussion in the section (??), we have

$$\Gamma(\pi) \simeq \text{Mor}_{\mathbf{H}/\mathbf{BU}(1)}(\pi; \bar{\rho}),$$

such that we could apply Γ in the previous diagram to get the following cospans in non-abelian twisted cohomology:

***** DIAGRAM *****

But recall that the previous space of section is *twisted nonabelian cohomology* and we need to work in *twisted generalized cohomology*. To pass from the nonabelian flavor to the generalized flavor, we need replace the commutative ring $\mathbb{C}$ to some E_∞ -ring R . In order to do this, observe that we have a functor $G_1 : \mathbf{CoRing} \rightarrow \mathbf{Grp}$ assigning to any commutative ring its group of units. For any R we have a canonical action $G_1(R) \times R \rightarrow R$ given by multiplication. Particularly, $G_1(\mathbb{C}) = U(1)$ so that the associated bundle obtained above a particular case of those $G_1(R)$ -bundles associated to the

Yoga

In order to build (co) quantizations following the previous recipes, we need to work in contexts in which we have the notion of fiber integration. Here we present some of such them. As we will see, in the standard cases they are given by some cohomology theory. There are some insights to look to such class of contexts. An almost trivial insight is that, as required, cohomology is presumably contravariant. But there are also more strong motivations, as we pass to discuss.

Observe that the main idea to build the integration $\int_{(-)} : \mathbf{H}^{op} \rightarrow \mathbf{D}$ is to suppose that any $H(X) \in \mathbf{D}$ is *dualizable*. This means that there is another $H(X)^\vee$ such that the the canonical maps

$$H(X) \otimes H(X)^\vee \rightarrow 1 \quad \text{and} \quad 1 \rightarrow H(X) \otimes H(X)^\vee$$

satisfying suitable commutative conditions, in the sense that the following triangle conditions:

*****DIAGRAM*****

Under such conditions, any morphism

$$f : H(X) \rightarrow H(Y) \quad \text{produces another} \quad f^\vee : H(Y)^\vee \rightarrow H(X)^\vee$$

in a functorial way, meaning that the dual of a composition is the composition of the dual maps and that the dual of the identity is the proper identity. We will suppose further that any of such objects is *self-dual* in the sense that we have (non necessarily unique) isomorphisms $H(X) \simeq H(X)^\vee$. Now, having selected one of such isomorphisms, for any $f : X \rightarrow Y$ we can consider its fiber integration as being the following composition:

$$H(X) \xrightarrow{\simeq} H(X)^\vee \xrightarrow{H(f)^\vee} H(Y)^\vee \xrightarrow{\simeq} H(Y)$$

$\int_f$

Now, we list some strong insights in the direction of considering pull-push contexts as being given by multiplicative cohomology theories:

1. the archetypal example of category in which any object is dualizable is the category of vector spaces over some field. Indeed,..... Generally, cohomology theories takes values in $\mathbf{Vec}_{\mathbb{K}}$;
2. we have a natural dual concept of cohomology groups: the homology groups;
3. when we are working with *orientable* manifolds, the choice of an orientation induces an isomorphism between cohomology and homology given by Poincaré-duality. Here we recall that
4. thinking homology as being the dual of cohomology, for any f we can take $H(f)^\vee$ as being the canonical map induce from f on homology;
5. the functor H need to be monoidal. Recall Kunnet theorem say that H is monoidal..... Then the functor H take values in the category of graded algebras monoidally structured by the graded tensor product: if $A_* = \oplus_i A_i$ and $B_* = \oplus_j B_j$ are graded algebras, we define $A^* \otimes B^*$ as being the graded vector space

$$(A^* \otimes B^*)_k = \bigoplus_{i+j=k} A_i \otimes B_j$$

with the product defined on homogeneous elements by

$$(a \otimes b) \cdot (a' \otimes b') = (-1)^{\deg a' \cdot \deg b} (a \otimes a') \cdot (b \otimes b')$$

and extended by linearity. The unity object is the field $\mathbb{K}$ viewed as a trivial graded algebra. Furthermore, as we are working with coefficients in a field, the Künneth formula (see, for instance, [??]) reduces to

$$H(X \times Y)^k \simeq \bigoplus_{i+j=k} H(X)^i \otimes H(Y)^j,$$

saying precisely that H is monoidal.

6. compatibility condition and naturality property are directly satisfied
 7. ziz-zag property and the six operations yoga of Grothendieck
- Problem: need that the phase space be smooth manifolds. But in this case we can give a precise definition of path integral, so that we do not have gained generality.
 - another problem: in the context of manifolds, if we consider De Rham cohomology we are forgetting possibly additional geometry structures.
 - But observe that, in the case that $\mathbf{Fields}(-) = \mathbf{H}(-, \mathbf{Fields})$, the fields functor fields is itself a cohomology, so there is a natural cohomology to consider. Such cohomology, as was discussed in the section ??, take into account geometric freatures.

*****DIAGRAM*****

Now, suppose that the context $\mathbf{H}$ has the notions of pullback and pushforward. More precisely, let is suppose that we have been chosen a monoidal functor $H : \mathbf{H}^{op} \rightarrow \mathbf{D}$ as well as other functors

$$(-)^* : \mathbf{H}^{op} \rightarrow \mathbf{D} \quad \text{and} \quad \int_{(-)} : \mathbf{H} \rightarrow \mathbf{D}$$

which coincide with H at objects. Then for any classical theory determined by the action S , we can try to build a quantum theory $\mathcal{Q}_S$ taking

$$\mathcal{Q}_S(\Sigma_0) = H(\mathbf{Fields}(\Sigma_0)) \quad \text{and} \quad \mathcal{Q}_S(\Sigma)(\psi) = \int_{H(j_1)} H(j_0)^*(\psi),$$

which means that $\mathcal{Q}_S(\Sigma)$ is simply the composition of the following dotted arrows:

We observe that, thanks to the monoidal property of the fields functor, the condition (5.1) is immediately satisfied. Now, in order to $\mathcal{Q}_S$ be a genuine quantum theory, it must be a functor. That is, it must preserve identities and compositions. The preservation of identities means that, for each cobordism $\Sigma : \Sigma_0 \rightarrow \Sigma_0$ diffeomorphic to the cylinder $\Sigma_0 \times I$, the above diagram is commutative. This can be obtained requiring that, for any $f : X \rightarrow Y$ on $\mathbf{H}$, we have $\int_f f^* = id_{H(X)}$. This seems to be very natural, because fiber integration is like a pushforward and, in general, we have that pullback and pushforward are dual notions. We will refer to this property saying that the pullback $(-)^*$ and the fiber integration $\int_{(-)}$ are *compatible*.

***** DIAGRAM *****

Let is analyze the preservation of compositions. So, given cobordisms $\Sigma : \Sigma_0 \rightarrow \Sigma_1$ and $\Sigma' : \Sigma_1 \rightarrow \Sigma_2$, let is look to the first below diagram. Applying $\mathbf{Fields}(-)$ and H we get

the second diagram. Observe that $\mathcal{Q}_S(\Sigma' \circ \Sigma)$ is obtained composing the dotted arrows, as well as $\mathcal{Q}_S(\Sigma') \circ \mathcal{Q}_S(\Sigma)$ is the composition of the punctured arrows. Therefore, the preservation of composition can be obtained requiring that, for any diagram in $\mathbf{H}$ like the first, the composition of dotted arrows coincide with the composition of the punctured arrows. We will refer to this saying that $(-)^*$ and $\int_{(-)}$ satisfy the *zig-zag property*.

At this moment, the conclusion is that, *if the notions of pullback and fiber integration are compatible and satisfy the zig-zag property, then really is a quantum theory*. But we are looking to a genuine quantization. So, we need to extend the rule $\mathcal{Q}_-$ to a functor. As we will see now, this can be done under suitable conditions on the pullback and on the fiber integration. So, given a morphism between classical theories $f_- : \mathbf{Fields} \rightarrow \mathbf{Fields}'$, to any Σ_0 we need assign a map

$$\mathcal{Q}_{f_{\Sigma_0}} : H(\mathbf{Fields}(\Sigma_0)) \rightarrow H(\mathbf{Fields}'(\Sigma_0))$$

making commutative the first dotted diagram. But recall that any f_{Σ} preserves the covariant phase space, so that we have a map $f_{\Sigma_0} : \mathbf{Fields}(\Sigma_0) \rightarrow \mathbf{Fields}'(\Sigma_0)$ and, therefore, a canonical map

$$H(f_{\Sigma_0}) : H(\mathbf{Fields}'(\Sigma_0)) \rightarrow H(\mathbf{Fields}(\Sigma_0)).$$

So, the main idea is to realize $\mathcal{Q}_-^{cov}$ not as a functor, but as a *contravariant* functor (i.e, as a *coquantization*).

dfs

The above discussion motivate the following definition: a *pull-push context* is given by a monoidal functor $H : \mathbf{H}^{op} \rightarrow \mathbf{D}$, defined on a concrete cartesian monoidal category $\mathbf{H}$, together other two functors

$$(-)^* : \mathbf{H}^{op} \rightarrow \mathbf{D} \quad \text{and} \quad \int_{(-)} : \mathbf{H} \rightarrow \mathbf{D},$$

respectively called *pullback* and *fiber integration* on H , which satisfy these conditions:

1. they coincide with H at objects;
2. they are compatible;
3. they have the zig-zag property;
4. they are natural.

Then, we have been proven the following result:

Theorem 5.3.1. *Let $H : \mathbf{H}^{op} \rightarrow \mathbf{D}$ be a pull-push context. Then for any category of classical theories $\mathcal{C} \subset \mathbf{H}$ there is a coquantization $\mathcal{Q}$ defined on the subcategory $D(\mathcal{Q}) \subset \mathcal{C}$ in which the covariant phase space maps $\mathbf{Fields}(-)$ extends to monoidal functors.*

Prequantization

Redefinition

Bibliography

- [1] J.-Freyd, T., The formal path integral and quantum mechanics, arXiv:1004.430
- [2] van Hove, L. , Mem. Acad. Roy. de Belgique, Classe des Sci. 26 (1951) No. 6.
- [3] Groenewold, H. J., *On the Principles of elementary quantum mechanics*, Physica 12, pp. 405-460, 1946
- [4] Gotay, M. J., <http://arxiv.org/abs/math-ph/9809011>
- [5] Gotay, M. J., Grabowski, J., Grundling, H. B., <http://arxiv.org/abs/dg-ga/9706001>
- [6] Gotay, M. J., Grundling, H. B., <http://arxiv.org/abs/dg-ga/9710024>
- [7] Gotay, M. J., *Functorial Geometric Quantization and Van Hove's Theorem*, Int. J. Th. Ph., vol. 19, n. 2, 1980.
- [8] Bartlett, B. H., *Categorical Aspects of Topological Quantum Field Theories*, arXiv:math/0512103.
- [9] Ayala, D., *Geometric Cobordism Categories*, arXiv:0811.2280
- [10] Segal, G., *The definition of conformal field theory*, Differential geometrical methods in theoretical physics (Como, 1987), NATO Adv. Sci. Inst. Ser. C Math. Phys. Sci., 250, Kluwer Acad. Publ., Dordrecht, (1988), 165-171
- [11] Baez, J. C., *Quantum Quandaries: a Category-Theoretic Perspective*, arXiv:quant-ph/0404040.
- [12] de Melo, W., de Faria, E., *Mathematical Aspects of Quantum Field Theory*, Cambridge University Press, 2010.
- [13] Lurie, J., *On the Classification of Topological Field Theories*, arXiv:0905.046.
- [14] Kelly, M., *Basic Concepts of Enriched Category Theory*, Cambridge University Press, 1982. Republished in: *Reprints in Theory and Applications of Categories*, No. 10 (2005) pp. 1-136.

- [15] Aguiar, M., Mahajan, S., *Hopf Algebras and Tensor Categories*, AMS, 2013.
- [16] Aguiar, M., Mahajan, S., *Monoidal Functors, Species and Hopf Algebras*, AMS, 2010.
- [17] Lurie, J., *Higher Algebra*.
- [18] Baez, J., Dolan, J., *Higher Algebra III: n-Categories and the Algebra of Opetopes*, arXiv:q-alg/9702014.
- [19] Mac Lane, S., *Categories for the Working Mathematician*, Springer, 1998.
- [20] Labastida, J., Marino, M., *Topological Quantum Field Theory and Four Manifolds*, Springer, 2005.
- [21] Ohtsuki, T., *Quantum Invariants*, World Scientific, 2002.
- [22] Paul S. Aspinwall Bridgeland, T., Craw, A., Douglas, M. R., Gross, M., Kapustin, A., Moore, G. W., Segal, G., Szendrői, B., Wilson, P. M. H., *Dirichlet Branes and Mirror Symmetry*, AMS, 2009.
- [23] Deligne, P., Etingof, P., Freed, D. S., Jeffrey, L. C., Kazhdan D., Morgan, J. W., Morrison, D. R., Witten, E., *Quantum Fields and Strings: A Course for Mathematicians*, vol. 1 and 2, AMS, 1999.
- [24] Sati, H., Schreiber, U. (editors), *Mathematical Foundations of Quantum Field Theory and Perturbative String Theory*, AMS, 2010.
- [25] Mariño, M., *Chern-Simons theory, matrix models and topological strings*, Oxford University Press, 2005.
- [26] Katz, S., *Enumerative Geometry and String Theory*, AMS, 2000.
- [27] Taubes, C. H., *Seiberg Witten and Gromov Invariants for Symplectic 4-Manifold*, International Press, 2000.
- [28] -Maluga, T. A., Brans, C., *Exotic Smoothness and Physics*, World Scientific, 2007.